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Boundary Limited Observer Systems

Empirical Equivalence, Interface Refinement, and the Limits of Latent Reconstruction

Abstract. An observation map can preserve every recorded output while failing to determine the latent state that produced it. We study this distinction for a deterministic interface $\pi : X \to O$. Exact decoding requires a left inverse and therefore injectivity; quantities determined by the observation are constant on its fibers. At the model level, we define empirical equivalence as equality of reachable observation sets. For every model with a nonempty latent space, adjoining an unobserved Boolean coordinate gives an equivalent model without an exact decoder. If the original is exactly decodable, the two models have opposite decoder statuses. Interface refinement describes which additional distinctions a measurement can retain. Exhaustive finite calculations reproduce the hidden multiplicities, collision counts, and refinement maps, and agree with twelve archived output tables. The result concerns set based identifiability within the declared model class. It does not establish equality of probability laws, impossibility of reconstruction from temporal measurements, or physical existence of the added coordinate. The proofs and computational details are supplied in appendices and a companion supplement.

Keywords: observer interface; reachable observations; latent reconstruction; identifiability; quotient; interface refinement.

1. Introduction

Scientific measurements represent selected features of a system. A measurement can be accurate for its intended purpose while leaving several source states compatible with the same output. The resulting inverse problem depends on the observation map, the allowed model class, and the quantity to be reconstructed.

Identifiability theory asks which model properties observations determine. Statistical formulations compare probability laws, while dynamical observability uses outputs together with specified evolution and inputs (Rothenberg, 1971; Hermann and Krener, 1977). The present analysis isolates a simpler question: what follows from a deterministic observation map and the set of outputs it can produce? This restriction permits exact statements about fibers, reconstruction, and comparison of interfaces.

The main construction appends a coordinate that the observation map ignores. The extended model produces the same reachable observations, but distinct hidden labels prevent exact reconstruction. The construction proves nonuniqueness within a model class that permits such extensions. Additional dynamics, interventions, or physical constraints may exclude them. Related limits in learned latent representations likewise depend on the assumptions imposed on the model class (Locatello et al., 2019).

The contribution is a unified formalization of these set theoretic facts, their refinement relations, and their finite computational witnesses. It does not establish a general impossibility of self-observation or a theory of consciousness. Section 2 defines the model and principal equations. Section 3 states the main result. Sections 4 and 5 give the methods and findings, followed by their implications and conclusion. Detailed proofs, counterexamples, and the broader research program follow in the appendices.

2. Observation model and principal equations

An interface is a map $\pi : X \to O$ from a source state space to an observation space. Two states are observationally equivalent when their outputs agree:

$$x \sim_\pi y \iff \pi(x) = \pi(y), \qquad Q_\pi = X/\!\sim_\pi. \tag{1}$$

Each class is a fiber of π. The quotient Q_π is bijective with the reachable range $\pi[X]$. A quantity $f : X \to V$ is determined on that range precisely when it is constant on every fiber, equivalently when it factors uniquely through the quotient. This is a statement about set functions; it supplies no algorithm or error bound.

A globally exact decoder is a total map $d : O \to X$ satisfying

$$d \circ \pi = \mathrm{id}_X \quad \Longrightarrow \quad \pi \text{ is injective.} \tag{2}$$

When $X \neq \varnothing$, injectivity is also sufficient: invert π on its range and use one fixed state on unreachable observations. When $X = \varnothing$, a total decoder exists exactly when $O = \varnothing$. The definitions distinguish a decoder on all of O from one defined only on reachable observations.

A model $\mathsf{M} = (X_\mathsf{M}, \pi_\mathsf{M})$ has reachable set $\mathscr{O}(\mathsf{M}) = \pi_\mathsf{M}[X_\mathsf{M}]$. Models with the same observation codomain are compared by

$$\mathsf{M} \approx \mathsf{M}' \iff \mathscr{O}(\mathsf{M}) = \mathscr{O}(\mathsf{M}'). \tag{3}$$

Throughout this paper, *empirical equivalence* means this equality of reachable sets. No probability law is part of the definition. An admissible experiment is a fixed function $e : O \to R$ that processes an observation. Its possible outcomes are

$$\mathscr{R}_e(\mathsf{M}) = e[\mathscr{O}(\mathsf{M})], \qquad \mathsf{M} \approx \mathsf{M}' \implies \mathscr{R}_e(\mathsf{M}) = \mathscr{R}_e(\mathsf{M}'). \tag{4}$$

Such an experiment neither changes the source dynamics nor acquires another measurement. Interventions and temporal records require an enlarged model.

Interfaces on the same source space can retain different distinctions. A fine interface $\pi_F : X \to O_F$ refines $\pi_C : X \to O_C$ when a coarsening map c satisfies

$$\pi_C = c \circ \pi_F. \tag{5}$$

This relation is reflexive and transitive. It induces a surjection $[x]_F \mapsto [x]_C$ between observable quotients. If a pair is separated by F and merged by C, that quotient map is noninjective and reverse refinement is impossible. The proofs are in Appendices B to D.

3. Main result: equivalent outputs and nonunique latent models

The hidden extension makes the reconstruction limit explicit. For a model with $X_{\mathsf{M}} \neq \varnothing$, define

$$X_{\mathsf{M}^*} = X_{\mathsf{M}} \times \{0,1\}, \qquad \pi_{\mathsf{M}^*}(x,b) = \pi_{\mathsf{M}}(x), \qquad p(x,b) = x. \tag{6}$$

The map p is surjective and preserves observations. Distinct Boolean labels over the same base state produce a collision.

Theorem 3.1 (Hidden extension and reconstruction)**.** *For every model with a nonempty latent space, the extension in* (6) *has the same reachable set and the same possible outcomes under every fixed observation processing map e. It has no globally exact decoder. If the base has an exact decoder, the two models have different decoder statuses.*

The proof follows from two identities. Each base output is produced by $(x,0)$, and every extended output is a base output. Meanwhile, $(x,0)$ and $(x,1)$ are distinct states with the same observation. Equation (2) excludes a decoder. The complete construction and supporting results appear in Appendices G and I.

For $O \neq \varnothing$, compare the identity model (O, id_O) with its Boolean extension. Both reach all of O, but only the identity model has an exact decoder. Thus no function of the reachable set can determine decoder existence for every model in a class containing both. This does not mean that every equivalence class contains both statuses, nor that adjoining a hidden coordinate changes the status of an already nondecodable base.

The same limitation holds for a complete family of Boolean predicates on observations. A predicate records whether it succeeds at any reachable output. Agreement for all predicates is equivalent to reachable-set equality, because predicates testing equality to individual observations determine membership. Model properties descend to the empirical quotient exactly when they are constant on its classes. These statements are proved in Appendices F and H.

4. Methods

The study combines deductive proofs with exhaustive finite calculations. The calculations enumerate declared mathematical models; no biological measurements or physical parameter estimates enter the analysis. The reference code is `BoundaryObserver` at commit eb96b984ae999ee6cbc0aae33c2d501c46d9540f. The formal source remains unchanged in this revision.

The finite base has $X = O = \{0,1,2,3,4,5\}$ and $\pi(x) = x$. A comparison model observes x mod 3. Hidden extensions use $m \in \{1,2,3,5,8\}$ labels for each base state, with $\pi_m(x,h) = x$. For each reachable output, the program enumerates its fiber and calculates

$$C(\mathsf{M}) = \sum_{o \in \mathscr{O}(\mathsf{M})} \binom{|\pi_{\mathsf{M}}^{-1}(o)|}{2}, \qquad C(\mathsf{M}_m) = 3m(m-1). \tag{7}$$

The collision count measures unordered pairs with equal outputs. It is not a probability or an entropy.

Four processing maps test identity, a threshold at three, a three category partition, and width two compression. Three predicates test evenness, values at least three, and membership in $\{0,5\}$.

A separate eight state example compares identity, paired buckets, binary regions, and a relabeled identity. Every claimed finite relation is evaluated over its full declared domain.

The revision reruns the eighteen repository tests and six additional tests of the assumptions and counterexamples. It also regenerates all twelve CSV outputs and compares their bytes with the archived files. The repository records successful Lean CI for the pinned commit; a new Lean build is not claimed here. The added examples have explicit proofs in Appendix J and are not represented as newly checked Lean declarations. Commands, exact outputs, and declaration correspondence are supplied in the package and companion supplement.

5. Findings

The finite calculations reproduce the predicted separation between latent multiplicity and reachable outputs. The injective base has six singleton fibers and a decoder. Its Boolean extension has twelve states, six two-state fibers, and no decoder. The modulo three comparison has three reachable outputs, three collisions, and no decoder.

Table 1. Exhaustive hidden extension calculations. Each row reaches the same six observations.

m	Latent states	Reachable outputs	Fiber size	Collisions	Exact decoder
1	6	6	1	0	Yes
2	12	6	2	6	No
3	18	6	3	18	No
5	30	6	5	60	No
8	48	6	8	168	No

All twenty eight model and experiment comparisons preserve the corresponding outcome set. All twenty one existential predicate comparisons and all twenty one complete truth-table comparisons agree with the base. These three selected predicates illustrate invariance; the theorem concerning every predicate is established separately by proof.

Refinement retains eight, four, and two observable classes for identity, paired buckets, and binary regions. Identity to paired, paired to region, and identity to region respectively collapse four, eight, and twelve previously distinguished pairs. Each induced quotient map is surjective and noninjective. Relabeling preserves all eight classes and gives a bijection. Complete tables and figures appear in Appendix L and the supplement.

All twenty four tests pass. The twelve regenerated CSV files are byte identical to the archived outputs. The finite counts use exact enumeration and integer arithmetic. They have no integration step, sampling uncertainty, or asymptotic numerical convergence claim. Exhaustive agreement for these examples does not establish a result for every model; the general conclusions depend on the proofs.

6. Discussion

The result identifies a restriction on inference from a specified interface. If a quantity varies within an observation fiber, that observation alone cannot determine it exactly. A stronger reconstruction claim must state the assumptions that eliminate the ambiguity. Such assumptions may include

known dynamics, additional sensors, interventions, regularity, or a justified restriction of the latent model class.

Equality of reachable sets is weaker than statistical equivalence. Bernoulli laws with parameters $1/10$ and $9/10$ both permit the outputs zero and one, but assign different probabilities to them. The present definition omits that distinction. A statistical extension must compare induced probability laws or likelihoods rather than only possible outputs (Rothenberg, 1971).

An instantaneous collision also need not prevent reconstruction from time dependent observations. For the known system

$$\dot{x}_1 = x_2, \qquad \dot{x}_2 = 0, \qquad y = x_1, \tag{8}$$

one output leaves x_2 undetermined, while $y(0)$ and $y(T)$, for $T \neq 0$, determine both initial coordinates. Measurement error in the reconstructed second coordinate is amplified as $|T|$ becomes small. The derivation and error bound are in Appendix J. This distinction is consistent with dynamical observability theory (Hermann and Krener, 1977).

Neural applications require the same separation between source state, measurement, and reconstruction. Published grid-cell studies motivate analysis of how population geometry depends on the measurement interface (Gardner et al., 2022). A spatial tuning map, a temporal population trajectory, a topological reconstruction, and evidence of attraction are different objects. The present set theoretic result supplies no inference from toroidal geometry to an attracting invariant torus. That inference requires specified dynamics and evidence concerning invariance and stability.

The hidden coordinate is a mathematical witness permitted by the model class. Its construction does not demonstrate an unmeasured physical variable. Questions about embedded self-modeling, consciousness, or continuity across scales require additional definitions and observations. They are retained as a research program in Appendices O and P, with their status separated from the theorem.

7. Conclusion

A declared observation map determines a partition of source states. Exact reconstruction requires that the map preserve every source distinction, and reachable-set equality does not determine decoder existence across an unrestricted class of latent models. The Boolean extension supplies a constructive witness, while interface refinement describes how a richer measurement can preserve distinctions lost by a coarser one.

The finite calculations reproduce the stated counts and invariance relations. The scope of the result remains deterministic and set based. Probability laws, temporal measurements, physical constraints, and reconstruction error must be specified before applying it to a scientific system. The resulting practical question is which additional measurement or model assumption would separate the states relevant to the proposed inference.

A. Mathematical preliminaries and proof convention

A.1 Functions, images, and fibers

The notation for functions, images, fibers, composition, inverses, and equivalence relations follows standard set-theoretic usage (Halmos, 1974, Chs. 8–11). For a function $f : X \to Y$, the image of a subset $A \subseteq X$ is

$$f[A] = \{f(x) : x \in A\}.$$

The range is $\mathrm{im}(f) = f[X]$. The fiber over $y \in Y$ is

$$f^{-1}(y) = \{x \in X : f(x) = y\}.$$

The function is injective when $f(x) = f(x')$ implies $x = x'$. It is surjective when every $y \in Y$ has at least one preimage.

A function $g : Y \to X$ is a left inverse of f when

$$g \circ f = \mathrm{id}_X .$$

A left inverse can exist only when f is injective (Mac Lane and Birkhoff, 1999, Chs. 1 and 4). This elementary fact is the exact-decoding obstruction used throughout the paper.

A.2 Equivalence relations and quotients

An equivalence relation partitions a set into disjoint equivalence classes. The quotient $X/\!\sim$ is the set of those classes. The canonical class map is

$$q : X \to X/\!\sim, \qquad q(x) = [x].$$

A function on X descends to the quotient precisely when it is constant on each equivalence class. The descended function is then unique because every quotient class has a representative. This is the elementary set-level form of a quotient universal property (Mac Lane and Birkhoff, 1999; Leinster, 2014).

A.3 State-level and model-level notation

The symbol X denotes a source or world state space. The symbol O denotes an accessible observation space. The observation map is $\pi : X \to O$. A generative model is written

$$\mathsf{M} = (X_{\mathsf{M}}, \pi_{\mathsf{M}}).$$

Its reachable observation set is $\mathscr{O}(\mathsf{M})$. The possible outcome set of an admissible experiment e is $\mathscr{R}_e(\mathsf{M})$.

Two equivalence relations are used. State-level observational equivalence identifies $x, y \in X$ when $\pi(x) = \pi(y)$. Model-level empirical equivalence identifies M and M' when $\mathscr{O}(\mathsf{M}) = \mathscr{O}(\mathsf{M}')$. The levels are not interchangeable.

A.4 Logical status of statements

Definitions fix the objects and relations used in the analysis. Propositions, lemmas, corollaries, and theorems are established by the proofs that follow them. Finite computations evaluate explicitly

enumerated instances of those definitions. Interpretations, hypotheses, applications, and future research questions are stated separately and do not inherit proof from the mathematical results. Literature citations identify established precedent, comparison, or empirical context.

B. Observer interfaces, accessible quantities, and exact decoding

The observer interface determines the distinctions available to every downstream quantity or test. The construction is deliberately static: unlike dynamical observability, which uses an output trajectory together with specified dynamics and inputs, it studies one declared map from source states to accessible observations (Kalman, 1963; Hermann and Krener, 1977; Sontag, 1998). This section develops the state level theory before introducing models.

Definition B.1 (Observer interface)**.** An observer interface on a world state space X is a pair

$$I = (O, \pi),$$

where O is an accessible observation space and $\pi : X \to O$ is the observation map.

Definition B.2 (Observational equivalence)**.** For $x, y \in X$, define

$$x \sim_I y \quad \Longleftrightarrow \quad \pi(x) = \pi(y).$$

Proposition B.3. *Observational equivalence is reflexive, symmetric, and transitive.*

Proof. Reflexivity follows from $\pi(x) = \pi(x)$. Symmetry follows because equality is symmetric. Transitivity follows because $\pi(x) = \pi(y)$ and $\pi(y) = \pi(z)$ imply $\pi(x) = \pi(z)$. □

The equivalence class of x is the fiber

$$[x]_I = \pi^{-1}(\pi(x)).$$

The class contains exactly the source states that the interface represents by the same accessible value. Equivalence is therefore relative to I. A second interface may separate states that the first interface merges.

B.1 Quantities that factor through the interface

Here accessibility means determination by a set function of the observation. Existence of that function does not establish an effective algorithm, finite computational cost, or stability under measurement error.

Definition B.4 (Interface accessible quantity)**.** Let $q : X \to V$. The quantity q factors through I when there exists a map $\bar{q} : O \to V$ such that

$$q = \bar{q} \circ \pi.$$

Theorem B.5 (Fiber constancy)**.** *If q factors through I, then q is constant on every observation fiber. In particular,*

$$x \sim_I y \quad \Longrightarrow \quad q(x) = q(y).$$

Proof. Assume $q = \bar{q} \circ \pi$. Let $x \sim_I y$. By definition, $\pi(x) = \pi(y)$. Applying $\bar{q}$ to both sides gives

$$\bar{q}(\pi(x)) = \bar{q}(\pi(y)).$$

Substitution of $q = \bar{q} \circ \pi$ yields $q(x) = q(y)$. □

The converse is developed through the observable quotient in Appendix C. A quantity constant on every fiber has a unique quotient representation.

Corollary B.6 (Variation prevents interface factorization)**.** *If $x \sim_I y$ and $q(x) \neq q(y)$, then q does not factor through I.*

Proof. If q factored through I, fiber constancy would imply $q(x) = q(y)$, contradicting the assumption. □

B.2 Internal tests

An internal test is a special case of an accessible quantity.

Definition B.7 (Internal test)**.** An internal binary test is a function

$$t : O \to \{\text{false}, \text{true}\}.$$

Proposition B.8 (No internal test separates one fiber)**.** *If $x \sim_I y$, then*

$$t(\pi(x)) = t(\pi(y))$$

for every internal test t.

Proof. The equality $\pi(x) = \pi(y)$ remains equal after applying the function t. □

This result is elementary, but it identifies the relevant restriction. The test is not weak because it is Boolean. It is restricted because its domain is O. Replacing the Boolean codomain by an arbitrary set does not restore distinctions that have already collapsed in the input.

B.3 Exact reconstruction

Exact reconstruction requires a left inverse of the observation map. Left inverses and injectivity are standard function-theoretic notions (Halmos, 1974, Chs. 10–11); their use here is specialized to an observer interface.

Definition B.9 (Globally exact decoder)**.** A globally exact decoder for I is a function $d : O \to X$ such that

$$d \circ \pi = \mathrm{id}_X .$$

Theorem B.10 (Exact decoding implies injectivity)**.** *If a globally exact decoder exists, then π is injective.*

Proof. Assume $d \circ \pi = \mathrm{id}_X$. Let $x, y \in X$ and assume $\pi(x) = \pi(y)$. Applying d gives

$$d(\pi(x)) = d(\pi(y)).$$

The left inverse condition reduces this equality to $x = y$. Therefore π is injective. □

Corollary B.11 (Noninjectivity prevents exact decoding)**.** *If π is noninjective, then no globally exact decoder exists.*

Proof. The statement is the contrapositive of Theorem B.10. □

The conclusion concerns exact reconstruction of every state. It does not prevent partial reconstruction, probabilistic inference, recovery under additional assumptions, or recovery after refining the interface. It states that one many to one map cannot have a left inverse on its entire domain.

B.4 Finite capacity obstruction

Finite interfaces make the obstruction quantitative. The proof uses only the finite cardinality obstruction for an injective map; no stochastic or numerical assumption is introduced.

Theorem B.12 (Finite pigeonhole obstruction)**.** *Let X and O be finite. If*

$$|O| < |X|,$$

then every observation map $\pi : X \to O$ is noninjective. Therefore there exist distinct $x, y \in X$ such that $x \sim_I y$, and no globally exact decoder exists.

Proof. Assume for contradiction that π is injective. An injective map from a finite set X into a finite set O implies $|X| \leq |O|$. This contradicts $|O| < |X|$. Hence π is noninjective. The collision and decoder conclusions follow from the definitions and the preceding corollary. □

The theorem does not require noise, approximation, or numerical instability. It is a cardinality result. If the interface has fewer accessible states than the source space, some source distinction must collapse.

C. The observable quotient and universal factorization

The observation map induces a canonical representation of the distinctions retained by the interface. The representation is the quotient of X by observational equivalence. Quotient sets and their representative-independent maps are standard constructions (Halmos, 1974, Chs. 8–11); the factorization language used below follows the universal-property viewpoint (Mac Lane and Birkhoff, 1999; Leinster, 2014).

Definition C.1 (Observable quotient)**.** The observable quotient of X relative to I is

$$Q_I = X/\!\sim_I.$$

The canonical class map is

$$q_I : X \to Q_I, \qquad q_I(x) = [x]_I.$$

The quotient discards the identities of states inside each fiber and retains the partition induced by observation. It is neither larger nor smaller than the available distinctions in the following sense.

Proposition C.2 (Quotient equality)**.** *For $x, y \in X$,*

$$q_I(x) = q_I(y) \quad \Longleftrightarrow \quad x \sim_I y.$$

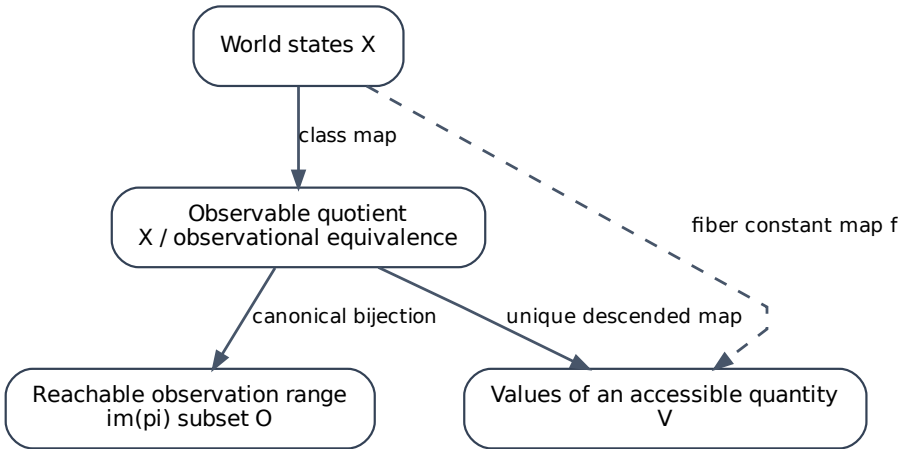


Figure 1. Universal factorization through the observable quotient. A quantity defined on world states descends uniquely to the quotient exactly when it is constant on observation fibers. The quotient is canonically equivalent to the reachable observation range.

Proof. Two quotient classes are equal exactly when their representatives are related by the defining equivalence relation. Here that relation is equality of observations. □

Let $\mathrm{im}(\pi) \subseteq O$ denote the reachable observation range. Each quotient class has one observation value, and each reachable observation value has one quotient class.

Theorem C.3 (Canonical equivalence with the observation range)**.** *The map*

$$\Phi : Q_I \to \mathrm{im}(\pi), \qquad \Phi([x]_I) = \pi(x),$$

is well defined and bijective.

Proof. Well definedness follows because representatives of one class have equal observations. Surjectivity follows from the definition of $\mathrm{im}(\pi)$. For injectivity, suppose $\Phi([x]_I) = \Phi([y]_I)$. Then $\pi(x) = \pi(y)$, so $x \sim_I y$ and $[x]_I = [y]_I$. □

The quotient and the reachable range are therefore equivalent descriptions of accessible resolution. The quotient emphasizes which source states have been identified. The range emphasizes which observation values remain reachable.

C.1 Universal property

The quotient is characterized by a universal property. Universal properties identify an object by the unique maps that factor through it rather than by a choice of representatives (Leinster, 2014).

Theorem C.4 (Observable quotient universal property)**.** *Let $f : X \to V$. The following statements are equivalent.*

(i) $f(x) = f(y)$ *whenever* $x \sim_I y$.
(ii) There exists a unique map $\bar{f} : Q_I \to V$ *such that*

$$f = \bar{f} \circ q_I.$$

Proof. Assume (i). Define $\bar{f}([x]_I) = f(x)$. The definition is independent of the representative because f is constant on each equivalence class. For every $x \in X$,

$$(\bar{f} \circ q_I)(x) = \bar{f}([x]_I) = f(x),$$

so the factorization exists. If $g : Q_I \to V$ also satisfies $f = g \circ q_I$, then for every class $[x]_I$,

$$g([x]_I) = g(q_I(x)) = f(x) = \bar{f}([x]_I).$$

Thus $g = \bar{f}$, proving uniqueness.

Conversely, assume (ii) and let $x \sim_I y$. Then $q_I(x) = q_I(y)$. Therefore

$$f(x) = \bar{f}(q_I(x)) = \bar{f}(q_I(y)) = f(y).$$

Hence f is constant on each fiber. □

The theorem provides a precise meaning for "observer accessible." A quantity is determined on reachable observations exactly when it is a function of the observable equivalence class. Any quantity that varies inside a class requires information not present in that quotient.

C.2 Composition with the observation range

The original definition of interface factorization used a map $\hat{f} : O \to V$. The quotient theorem uses a map $\bar{f} : Q_I \to V$. These formulations agree on reachable observations. Extension to all of O is automatic when $V \neq \varnothing$, by choosing a value on unreachable observations. If $V = \varnothing$, a map $X \to V$ requires $X = \varnothing$; a map $O \to V$ then exists only if $O = \varnothing$. A function on the full codomain O may assign arbitrary values to unreachable observations, while a function on Q_I contains only empirically realized classes. The distinction becomes useful at the model level, where the empirical quotient is represented exactly by realizable reachable sets.

D. Interface refinement and information ordering

Different instruments or representational procedures may observe the same world at different resolutions. Interface refinement is a preorder on interfaces. Identifying mutually refining interfaces gives an order on the corresponding classes. This resembles comparison of statistical experiments by informativeness, but the present relation is deterministic and is defined by an explicit commuting coarsening map rather than by a Markov kernel or decision-risk criterion (Blackwell, 1951, 1953; Torgersen, 1991).

Let

$$F = (O_F, \pi_F) \qquad \text{and} \qquad C = (O_C, \pi_C)$$

be interfaces on the same world state space X.

Definition D.1 (Interface refinement)**.** The interface F refines C, written

$$F \succeq C,$$

when there exists a coarsening map $c : O_F \to O_C$ such that

$$\pi_C = c \circ \pi_F.$$

The commuting equation states that every coarse observation is determined by a set function of the fine observation. It does not state that the fine observation can be recovered from the coarse observation.

Proposition D.2 (Refinement is reflexive and transitive)**.** *Every interface refines itself. If $F \succeq M$ and $M \succeq C$, then $F \succeq C$.*

Proof. Reflexivity uses the identity map on O_F. For transitivity, let $a : O_F \to O_M$ and $b : O_M \to O_C$ witness the two refinements. Then

$$\pi_C = b \circ \pi_M = b \circ a \circ \pi_F,$$

so $b \circ a$ witnesses $F \succeq C$. □

D.1 Monotonicity of accessible structure

Refinement orders observational equivalence in the expected direction.

Theorem D.3 (Fine equivalence implies coarse equivalence)**.** *If $F \succeq C$ and $x \sim_F y$, then $x \sim_C y$.*

Proof. Let c witness the refinement. The equality $\pi_F(x) = \pi_F(y)$ implies

$$c(\pi_F(x)) = c(\pi_F(y)).$$

Using $\pi_C = c \circ \pi_F$, this becomes $\pi_C(x) = \pi_C(y)$. □

The converse need not hold. A coarse interface may identify states that a fine interface distinguishes.

Theorem D.4 (Factorization is monotone under refinement)**.** *If $F \succeq C$ and $q : X \to V$ factors through C, then q factors through F.*

Proof. Let $q = \bar{q}_C \circ \pi_C$ and $\pi_C = c \circ \pi_F$. Then

$$q = \bar{q}_C \circ c \circ \pi_F.$$

Thus q factors through F by the map $\bar{q}_F = \bar{q}_C \circ c$. □

Every coarse test can therefore be simulated from the fine observation. Exact reconstruction also moves in the same direction.

Proposition D.5 (A coarse decoder induces a fine decoder)**.** *If $F \succeq C$ and C has a globally exact decoder, then F has a globally exact decoder.*

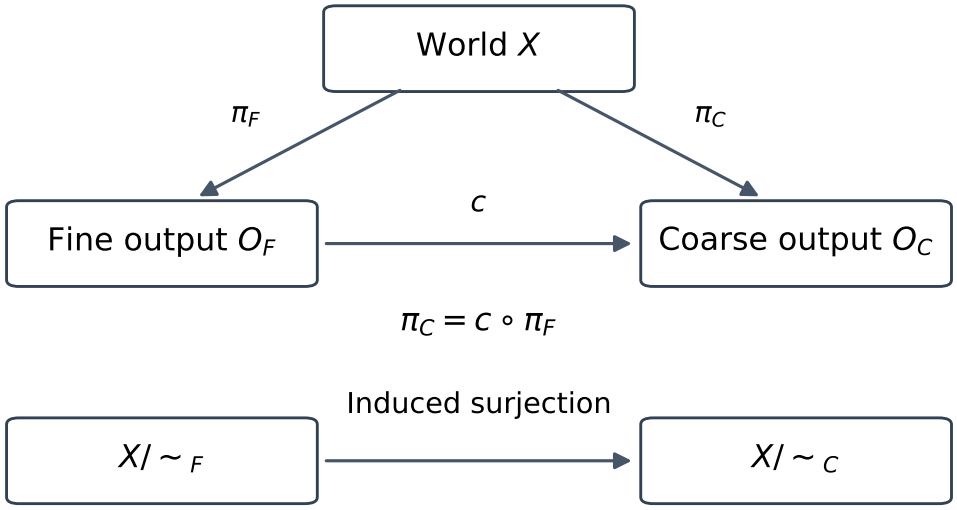


Figure 2. Interface refinement and induced quotient map. The coarsening map makes the observation triangle commute. The induced map sends each fine observational class to its coarse class. The map is always surjective. Under strict refinement it is not injective.

Proof. Let $d_C : O_C \to X$ satisfy $d_C \circ \pi_C = \mathrm{id}_X$, and let $c : O_F \to O_C$ witness refinement. Define $d_F = d_C \circ c$. Then

$$d_F \circ \pi_F = d_C \circ c \circ \pi_F = d_C \circ \pi_C = \mathrm{id}_X .$$

□

The contrapositive is useful: if the fine interface is noninjective, then no coarsening can be injective. Coarsening cannot restore a distinction that has already disappeared.

D.2 Refinement maps between observable quotients

Refinement induces a map between observable quotients.

Definition D.6 (Induced quotient map)**.** If $F \succeq C$, define

$$Q_{F\to C} : X/\!\sim_F \longrightarrow X/\!\sim_C$$

by

$$Q_{F\to C}([x]_F) = [x]_C.$$

The map is well defined because fine equivalence implies coarse equivalence.

Theorem D.7 (Surjectivity of the quotient map)**.** *For every refinement $F \succeq C$, the map $Q_{F\to C}$ is surjective.*

Proof. Let $[x]_C$ be any coarse class. The fine class $[x]_F$ maps to it by definition:

$$Q_{F\to C}([x]_F) = [x]_C.$$

Therefore every coarse class has a fine preimage. □

Definition D.8 (Strict refinement)**.** The refinement $F \succeq C$ is strict when there exist $x, y \in X$ such that

$$\pi_F(x) \neq \pi_F(y) \qquad \text{and} \qquad \pi_C(x) = \pi_C(y).$$

A strict refinement preserves at least one distinction that the coarse interface removes.

Theorem D.9 (Strict refinement makes the quotient map noninjective)**.** *If F strictly refines C, then $Q_{F\to C}$ is not injective.*

Proof. Choose x, y witnessing strictness. Since $\pi_F(x) \neq \pi_F(y)$, the fine classes are distinct:

$$[x]_F \neq [y]_F.$$

Since $\pi_C(x) = \pi_C(y)$, the coarse classes are equal. Hence

$$Q_{F\to C}([x]_F) = [x]_C = [y]_C = Q_{F\to C}([y]_F),$$

so the quotient map is noninjective. □

Theorem D.10 (Strict refinement prevents reverse refinement)**.** *If F strictly refines C, then C does not refine F.*

Proof. Suppose that $C \succeq F$. Fine equivalence would then follow from coarse equivalence. The strictness witnesses satisfy $x \sim_C y$ but $x \not\sim_F y$, which is a contradiction. □

D.3 Equal observational resolution

Two interfaces can use different observation labels while preserving exactly the same partition of X.

Definition D.11 (Same observational resolution)**.** Interfaces I and J have the same observational resolution when

$$x \sim_I y \quad \Longleftrightarrow \quad x \sim_J y$$

for every $x, y \in X$.

Mutual refinement implies equal observational resolution. Conversely, on reachable observations, equal resolution supplies mutually compatible class maps. The induced quotient maps are bijections.

Theorem D.12 (Mutual refinement gives quotient equivalence)**.** *If $I \succeq J$ and $J \succeq I$, then the induced quotient maps are inverse bijections. Thus the two interfaces retain the same observational distinctions even when their observation representations differ.*

This distinction separates strict coarsening from relabeling. Strict coarsening changes the partition of the world. Relabeling changes the symbols used for the classes but preserves the classes themselves.

E. Generative models and admissible experiments

The state level framework extends to models with different latent spaces. The common comparison object is the accessible observation space. Comparing models by the outputs they can generate is adjacent to structural identifiability and statistical experiment theory, although the present definition uses equality of reachable sets rather than equality of distributions or parameterized trajectories (Rothenberg, 1971; Walter and Pronzato, 1997; Blackwell, 1951; Torgersen, 1991).

Definition E.1 (Generative model)**.** A generative model over an observation space O is a pair

$$\mathsf{M} = (X_{\mathsf{M}}, \pi_{\mathsf{M}}),$$

where X_{M} is the latent state space and $\pi_{\mathsf{M}} : X_{\mathsf{M}} \to O$ is the observation map.

Definition E.2 (Reachable observation set)**.** The reachable observation set of M is

$$\mathscr{O}(\mathsf{M}) = \{\pi_{\mathsf{M}}(x) : x \in X_{\mathsf{M}}\} = im(\pi_{\mathsf{M}}).$$

Definition E.3 (Empirical equivalence)**.** Two models M and M' are empirically equivalent when

$$\mathsf{M} \approx \mathsf{M}' \quad \Longleftrightarrow \quad \mathscr{O}(\mathsf{M}) = \mathscr{O}(\mathsf{M}').$$

Empirical equivalence is reflexive, symmetric, and transitive because set equality has those properties. The relation compares what the models can produce at the specified observation boundary. It does not compare latent identity directly.

E.1 Observation only experiments

An admissible experiment is intentionally restricted to the accessible observation. This follows the general methodological separation between an unobserved model state and the experiment or decision rule applied to its observable output (Blackwell, 1953; Le Cam, 1986; Le Cam and Yang, 2000).

Definition E.4 (Admissible experiment)**.** An admissible experiment is a function

$$e : O \to R.$$

Its input is an accessible observation. It does not receive a latent state.

Definition E.5 (Possible outcome set)**.** The possible outcome set of e under M is

$$\mathscr{R}_e(\mathsf{M}) = \{e(\pi_{\mathsf{M}}(x)) : x \in X_{\mathsf{M}}\} = e[\mathscr{O}(\mathsf{M})].$$

Theorem E.6 (Empirical equivalence preserves all admissible outcomes)**.** *If* $\mathsf{M} \approx \mathsf{M}'$*, then*

$$\mathscr{R}_e(\mathsf{M}) = \mathscr{R}_e(\mathsf{M}')$$

for every admissible experiment $e : O \to R$*.*

Proof. Empirical equivalence gives $\mathscr{O}(\mathsf{M}) = \mathscr{O}(\mathsf{M}')$. Applying the same function e to equal sets gives

$$e[\mathscr{O}(\mathsf{M})] = e[\mathscr{O}(\mathsf{M}')].$$

These images are the two possible outcome sets. □

The theorem applies to arbitrary result types. A Boolean test, categorical classifier, numerical compression, and identity map are all admissible experiments when their domain is O.

Definition E.7 (Experimental separation)**.** An experiment separates M and M' when

$$\mathscr{R}_e(\mathsf{M}) \neq \mathscr{R}_e(\mathsf{M}').$$

Corollary E.8 (No admissible separation inside one empirical class)**.** *If* $\mathsf{M} \approx \mathsf{M}'$*, no admissible experiment separates the models.*

E.2 The identity experiment

The identity experiment retains the complete accessible observation value.

Definition E.9 (Identity experiment)**.** Define

$$e_{\mathrm{id}} : O \to O, \qquad e_{\mathrm{id}}(o) = o.$$

Proposition E.10. *For every model* M*,*

$$\mathscr{R}_{e_{\mathrm{id}}}(\mathsf{M}) = \mathscr{O}(\mathsf{M}).$$

Proof. The image of $\mathscr{O}(\mathsf{M})$ under the identity function is $\mathscr{O}(\mathsf{M})$ itself. □

The identity experiment is useful because it rules out an explanation based only on downstream compression. Even when the experiment returns every reachable accessible observation unchanged, the observation set may still fail to determine latent cardinality, fiber structure, or exact reconstructability.

E.3 Properties determined by experiments

Let P be a property of generative models. The property is determined by a fixed experiment e when there exists a criterion C on possible outcome sets such that

$$P(\mathsf{M}) \quad \Longleftrightarrow \quad C(\mathscr{R}_e(\mathsf{M}))$$

for every model M. Any property determined in this way is invariant under empirical equivalence, because empirically equivalent models have equal outcome sets.

This observation becomes decisive after strict hidden extensions are introduced. For $O \neq \varnothing$, the identity model and its Boolean extension have different decoder statuses inside one empirical class. Therefore it cannot be determined by any admissible experiment, including the identity experiment.

F. Predicate profiles and empirical identifiability

A single experiment may omit accessible information. The complete family of accessible predicates removes that objection. The use of complete external tests to characterize an observational

equivalence has analogues in automata and process semantics (Moore, 1956; Park, 1981; Milner, 1989; Rutten, 2000); here the predicates range over the accessible observation space rather than over transition traces. This section shows that complete predicate agreement characterizes empirical equivalence, yet still does not determine exact latent reconstructability.

Definition F.1 (Predicate success)**.** Let $p : O \to \{\text{false}, \text{true}\}$. The predicate p can succeed under M when

$$\exists x \in X_{\mathsf{M}} \quad p(\pi_{\mathsf{M}}(x)) = \text{true}.$$

Equivalently, at least one reachable observation satisfies p.

Definition F.2 (Predicate profile)**.** The predicate profile of M is the function

$$\mathsf{PredProf}(\mathsf{M}) : p \longmapsto \big[p \text{ can succeed under } \mathsf{M}\big].$$

Theorem F.3 (Empirical equivalence implies predicate equivalence)**.** *If $\mathsf{M} \approx \mathsf{M}'$, then every accessible predicate succeeds under M exactly when it succeeds under M'.*

Proof. Let p succeed under M. Then some $o \in \mathscr{O}(\mathsf{M})$ satisfies $p(o)$. Empirical equivalence gives $o \in \mathscr{O}(\mathsf{M}')$, so p succeeds under M'. The reverse direction is symmetric. □

The converse uses predicates that test equality to one observation.

Theorem F.4 (Complete predicate agreement implies empirical equivalence)**.** *Suppose every predicate $p : O \to \{\text{false}, \text{true}\}$ succeeds under M exactly when it succeeds under M'. Then*

$$\mathscr{O}(\mathsf{M}) = \mathscr{O}(\mathsf{M}').$$

Proof. Let $o \in \mathscr{O}(\mathsf{M})$. Define the equality predicate

$$p_o(a) = \begin{cases} \text{true}, & a = o, \\ \text{false}, & a \neq o. \end{cases}$$

The predicate succeeds under M, because o is reachable. By the assumed predicate agreement, it also succeeds under M'. Thus $o \in \mathscr{O}(\mathsf{M}')$. This proves $\mathscr{O}(\mathsf{M}) \subseteq \mathscr{O}(\mathsf{M}')$. The reverse inclusion follows by symmetry. □

Corollary F.5 (Predicate profile characterization)**.** *Two models are empirically equivalent if and only if their complete predicate profiles are equal.*

The complete profile therefore contains exactly the information in the reachable observation set. It is complete at the empirical boundary. It does not contain the multiplicities or identities of latent preimages.

F.1 Observation invariant and identifiable properties

Identifiability is always relative to a declared observation model and model class (Rothenberg, 1971; Walter and Pronzato, 1997; Raue et al., 2009; Villaverde et al., 2016). The definitions below isolate the corresponding invariance condition for the present reachable-set equivalence.

Definition F.6 (Observation invariant property)**.** A model property P is observation invariant when

$$\mathsf{M} \approx \mathsf{M}' \quad \Longrightarrow \quad \big(P(\mathsf{M}) \Longleftrightarrow P(\mathsf{M}')\big).$$

Definition F.7 (Empirically identifiable property)**.** A property is empirically identifiable in the present set based framework when its truth value is constant on every empirical equivalence class.

The two definitions are equivalent. The terminology emphasizes two views of the same condition. Observation invariance is a relation between representatives. Empirical identifiability is a property of the quotient class.

Any property determined by reachable observations, a fixed admissible experiment, or the complete predicate profile is observation invariant. The proof is the same in each case: empirically equivalent models supply equal criterion inputs.

Theorem F.8 (Exact latent reconstructability is not determined by any admissible experiment)**.** *Assume $O \neq \varnothing$ and a model class containing the identity model (O, id_O) and its Boolean hidden extension. For any fixed admissible experiment $e : O \to R$, no criterion on $\mathscr{R}_e(\mathsf{M})$ determines exact decoder existence for each model in that class.*

Proof. A property determined by an admissible experiment must be observation invariant. Choose $o_0 \in O$ and use the identity model, which has the identity decoder. In Appendix G, an exactly decodable model is paired with an empirically equivalent strict hidden extension that is not exactly decodable. Exact latent reconstructability is therefore not observation invariant. Hence no admissible experiment determines it. □

Corollary F.9. *Under the same assumptions, exact latent reconstructability is not determined by any accessible binary test, by the identity experiment, by the full reachable observation set, or by the complete predicate profile.*

This is stronger than a failure of one selected diagnostic. The complete collection of accessible predicates determines the empirical class, and the property still varies inside that class.

G. Hidden extensions and latent covers

The preceding sections identify what empirical equivalence preserves. Related identifiability results show that latent representations require additional assumptions to become unique (Teicher, 1963; Allman et al., 2009; Locatello et al., 2019; Hyvärinen et al., 2019; Khemakhem et al., 2020); the construction below gives a deterministic set-level witness tailored to the present observer interface. This section constructs models that share all empirically preserved information while differing in latent structure.

G.1 Duplicate and hidden factor extensions

Let $\mathsf{M} = (X_{\mathsf{M}}, \pi_{\mathsf{M}})$ be a generative model.

Definition G.1 (Hidden factor extension)**.** For an inhabited set H, define the hidden factor extension M^H by

$$X_{\mathsf{M}^H} = X_{\mathsf{M}} \times H$$

and

$$\pi_{\mathsf{M}^H}(x,h) = \pi_{\mathsf{M}}(x).$$

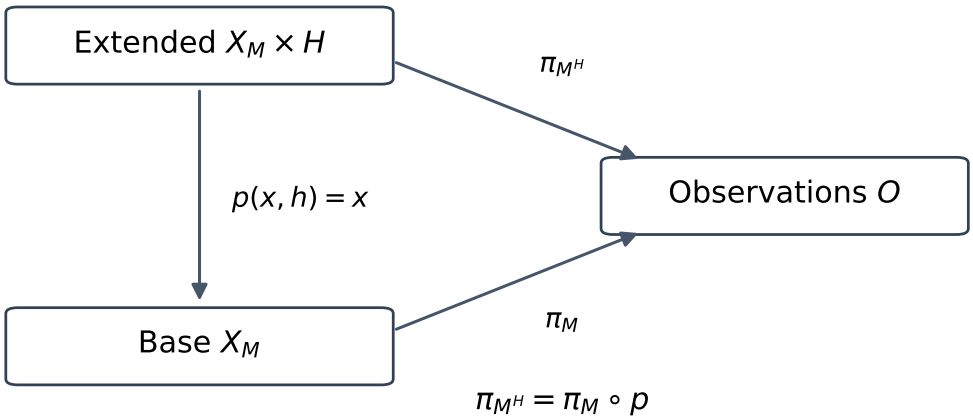


Figure 3. Hidden factor extension and observation preserving projection. The projection is surjective. Distinct hidden values attached to one base state create distinct extended states with one common observation.

The observation is independent of the hidden coordinate. Choosing a fixed anchor $h_0 \in H$ gives a surjective projection

$$p : X_{\mathsf{M}} \times H \to X_{\mathsf{M}}, \qquad p(x,h) = x.$$

The projection satisfies

$$\pi_{\mathsf{M}^H}(x,h) = \pi_{\mathsf{M}}(p(x,h)).$$

Theorem G.2 (Hidden extensions preserve empirical equivalence)**.** *For every inhabited H,*

$$\mathscr{O}(\mathsf{M}^H) = \mathscr{O}(\mathsf{M}).$$

Therefore $\mathsf{M}^H \approx \mathsf{M}$.

Proof. Every extended observation has the form $\pi_{\mathsf{M}^H}(x,h) = \pi_{\mathsf{M}}(x)$, so $\mathscr{O}(\mathsf{M}^H) \subseteq \mathscr{O}(\mathsf{M})$. Conversely, choose an anchor $h_0 \in H$. For every $x \in X_{\mathsf{M}}$, the extended state (x,h_0) produces $\pi_{\mathsf{M}}(x)$. Hence $\mathscr{O}(\mathsf{M}) \subseteq \mathscr{O}(\mathsf{M}^H)$. □

Corollary G.3 (Experimental invariance of hidden extensions)**.** *For every admissible experiment e,*

$$\mathscr{R}_e(\mathsf{M}^H) = \mathscr{R}_e(\mathsf{M}).$$

Theorem G.4 (Hidden collisions)**.** *If H contains distinct values $h_0 \neq h_1$ and X_{M} contains x, then*

$$(x,h_0) \neq (x,h_1)$$

but

$$\pi_{\mathsf{M}^H}(x,h_0) = \pi_{\mathsf{M}^H}(x,h_1).$$

Therefore the extended observation map is noninjective and M^H has no exact latent decoder.

Proof. The ordered pairs are distinct because their second coordinates differ. Both observations reduce by definition to $\pi_{\mathsf{M}}(x)$. The observation map is therefore noninjective. Exact latent decoding would imply injectivity, which is impossible. □

The Boolean extension uses $H = \{0,1\}$. The duplicate model is the same two copy construction written as a specific finite hidden factor.

G.2 Finite hidden multiplicity

For a positive integer m, define

$$X_{\mathsf{M}^{(m)}} = X_{\mathsf{M}} \times \{0,1,\dots,m-1\}$$

and ignore the second coordinate during observation. Each base latent state generates m extended states in one observation fiber.

Proposition G.5 (Arbitrary finite hidden families)**.** *For every base state $x \in X_{\mathsf{M}}$, the map*

$$\iota_x : \{0,\dots,m-1\} \to X_{\mathsf{M}^{(m)}}, \qquad \iota_x(j) = (x,j),$$

is injective, and every state in its image produces $\pi_{\mathsf{M}}(x)$.

Thus finite latent multiplicity can be increased without changing reachable observations or any admissible outcome set. In the verified source, `finiteMultiplicityModel M n` uses the hidden type $\mathrm{Fin}(n+1)$; the paper parameter m is therefore $n+1$.

G.3 Countable hidden multiplicity

The construction is not restricted to finite hidden sets. Let

$$X_{\mathsf{M}^{(\mathbb{N})}} = X_{\mathsf{M}} \times \mathbb{N}$$

and define

$$\pi_{\mathsf{M}^{(\mathbb{N})}}(x,n) = \pi_{\mathsf{M}}(x).$$

For each x, the map $n \mapsto (x,n)$ embeds $\mathbb{N}$ into the observation fiber over $\pi_{\mathsf{M}}(x)$. The model remains empirically equivalent to M, and every admissible outcome set remains equal. If $X_{\mathsf{M}} \neq \varnothing$, distinct natural number coordinates give a collision, so exact latent decoding is impossible.

This result separates empirical class size from latent fiber size. An empirical class can contain representatives with one, finitely many, or countably many latent states above the same accessible observation.

G.4 Latent covers

The product construction is one example of a more general relation.

Definition G.6 (Latent cover)**.** An extended model M^* is a latent cover of a base model M when there exists a surjective projection

$$p : X_{\mathsf{M}^*} \to X_{\mathsf{M}}$$

such that

$$\pi_{\mathsf{M}^*}(z) = \pi_{\mathsf{M}}(p(z))$$

for every $z \in X_{\mathsf{M}^*}$.

Latent covers compose. The identity map makes each model a cover of itself.

Theorem G.7 (A latent cover is empirically equivalent to its base)**.** *If* M^* *covers* M*, then*

$$\mathscr{O}(\mathsf{M}^*) = \mathscr{O}(\mathsf{M}).$$

Consequently, every admissible experiment has the same possible outcomes under both models.

Proof. Observation preservation shows that every observation of M^* is an observation of M. Surjectivity shows that every base latent state has an extended preimage, and observation preservation gives the same accessible output. Hence the reachable sets are equal. Outcome equality follows from Theorem E.6. □

Definition G.8 (Strict latent cover)**.** A latent cover is strict when its projection p is noninjective.

A strict cover contains distinct extended states that project to one base state. Observation preservation then forces those extended states to share one observation.

Theorem G.9 (Strict covers prevent exact latent reconstruction)**.** *If* M^* *is a strict latent cover of* M*, then* π_{M^*} *is noninjective and* M^* *has no exact latent decoder.*

Proof. Since the projection is noninjective, choose $z_1 \neq z_2$ with $p(z_1) = p(z_2)$. Observation preservation gives

$$\pi_{\mathsf{M}^*}(z_1) = \pi_{\mathsf{M}}(p(z_1)) = \pi_{\mathsf{M}}(p(z_2)) = \pi_{\mathsf{M}^*}(z_2).$$

Thus the extended observation map is noninjective. Exact decoding is therefore impossible. □

G.5 Existence of strict covers for inhabited models

Every inhabited model has a Boolean strict hidden extension. The construction can then be repeated.

Theorem G.10 (Every inhabited model admits a strict latent cover)**.** *For every generative model with at least one latent state, there exists an empirically equivalent strict latent cover with additional latent distinctions.*

Proof. Use the Boolean hidden extension. Projection onto the base state is surjective and observation preserving. The two Boolean values make the projection noninjective. □

The Lean predicate `LatentlyMaximal` means that no strict latent cover exists. The theorem negates that predicate for every inhabited model. This definition is not ordinary maximality in a partial order. A strict cover means a noninjective surjective projection; for infinite models such a projection may be a self-map. For example, $X = \mathbb{N}$, $O = \{0\}$, and $\pi(n) = 0$ admit the strict self-cover $p(n) = \lfloor n/2 \rfloor$. Surjectivity follows from $p(2n) = n$, and $p(0) = p(1)$ proves noninjectivity. Additional hidden labels also need not increase an infinite set's cardinality. The proved conclusion is the existence of strict covers within the specified model class. It does not assert a physically realized hierarchy of hidden variables.

H. The empirical model quotient

Empirical equivalence partitions generative models into classes with equal reachable observation sets. The empirical model quotient records only this accessible information. The descent results use the same representative-independence principle as ordinary quotient constructions and universal properties (Mac Lane and Birkhoff, 1999; Leinster, 2014), now applied to a space of models.

Definition H.1 (Empirical model quotient)**.** Let $\mathfrak{M}_O$ denote the collection of generative models over O, with a fixed set theoretic universe for latent spaces. The Lean development makes the corresponding universe parameters explicit. The empirical model quotient is

$$\mathfrak{M}_O/\approx,$$

where $\mathsf{M} \approx \mathsf{M}'$ means $\mathscr{O}(\mathsf{M}) = \mathscr{O}(\mathsf{M}')$.

The canonical projection sends each model to its empirical class:

$$q_{\mathscr{E}} : \mathfrak{M}_O \to \mathfrak{M}_O/\approx.$$

Two canonical classes are equal exactly when the corresponding models are empirically equivalent.

H.1 Invariant properties descend

Theorem H.2 (Universal property for model properties)**.** *A property $P : \mathfrak{M}_O \to \{\text{false}, \text{true}\}$ is observation invariant if and only if there exists a unique property*

$$\bar{P} : \mathfrak{M}_O/\approx \to \{\text{false}, \text{true}\}$$

such that

$$P = \bar{P} \circ q_{\mathscr{E}}.$$

Proof. If P is observation invariant, define $\bar{P}([\mathsf{M}]) = P(\mathsf{M})$. This is well defined because all representatives of one class have the same truth value. The factorization follows from the definition, and uniqueness follows because every quotient class has a representative. Conversely, any property factored through the quotient is constant on each quotient class and is therefore observation invariant. □

The same result holds for arbitrary valued measurements.

Theorem H.3 (Universal property for invariant measurements)**.** *Let $F : \mathfrak{M}_O \to V$. The map F assigns equal values to empirically equivalent models if and only if there exists a unique*

$$\bar{F} : \mathfrak{M}_O/\approx \to V$$

such that $F = \bar{F} \circ q_{\mathscr{E}}$.

H.2 Realizable observation sets

The quotient has a concrete representation.

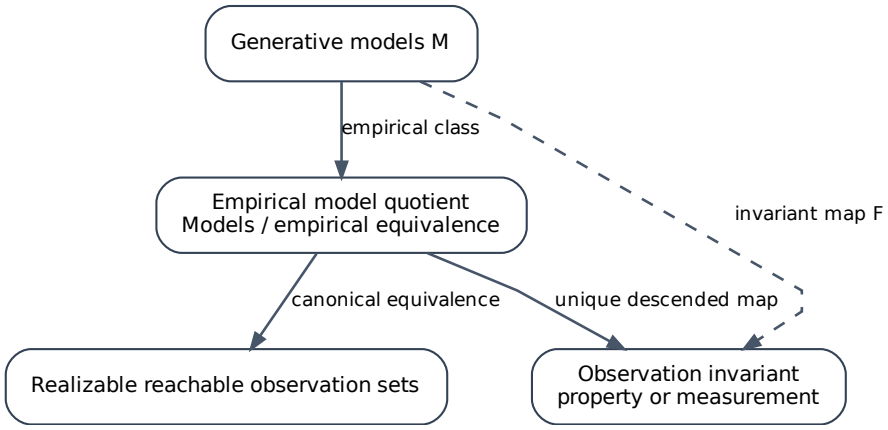


Figure 4. Empirical quotient and invariant descent. Observation invariant properties and measurements factor uniquely through empirical classes. Exact latent reconstructability fails this condition because it varies among representatives of one class.

Definition H.4 (Realizable observation set)**.** A subset $A \subseteq O$ is realizable when there exists a generative model M such that

$$\mathscr{O}(\mathsf{M}) = A.$$

Theorem H.5 (Canonical representation of empirical classes)**.** *The map*

$$[\mathsf{M}] \longmapsto \mathscr{O}(\mathsf{M})$$

is a bijection between the empirical model quotient and the collection of realizable reachable observation sets.

Proof. The map is well defined because empirically equivalent models have equal reachable sets. It is injective because equal reachable sets imply empirical equivalence and therefore equal quotient classes. It is surjective by the definition of realizability. □

This theorem specifies exactly what the empirical quotient retains. It retains the reachable set. It does not retain latent cardinality, observation fiber size, hidden factor type, or exact decoder status.

H.3 Exact reconstruction does not descend

Theorem H.6 (Exact latent reconstruction is not a quotient property)**.** *Assume $O \neq \varnothing$ and a model class containing (O, id_O) and its Boolean hidden extension. Exact latent reconstructability does not factor through the empirical model quotient of that class.*

Proof. Use the identity model on O; it is inhabited and has the identity decoder. Its Boolean hidden extension is empirically equivalent to the base. The extension has a noninjective observation map and no exact latent decoder. Therefore the property has different truth values on two representatives of one empirical class. A property with that variation cannot descend to the quotient. □

The theorem is the quotient form of Theorem F.8. The identity model and its Boolean extension have the same reachable set and opposite decoder statuses. Thus no criterion based on that set works for every model in the stated class. This is a global nonidentifiability result, not a claim that every empirical class contains both decoder statuses. In particular, an empty reachable set forces an empty latent space; its total decoder exists exactly when $O = \varnothing$.

I. Synthesis theorem

The preceding results combine into one constructive theorem. The theorem begins with an arbitrary inhabited model and an arbitrary admissible experiment. It returns a strict hidden extension that preserves the complete accessible boundary while preventing exact latent reconstruction.

Theorem I.1 (Boundary limited observer synthesis)**.** *Let* $\mathsf{M} = (X_{\mathsf{M}}, \pi_{\mathsf{M}})$ *be a generative model, let* $x_0 \in X_{\mathsf{M}}$ *be an inhabited witness, and let* $e : O \to R$ *be an admissible experiment. Then there exists a model* M^* *such that:*

(i) M^* *is a strict latent cover of* M*;*
(ii) the empirical classes are equal, $q_{\mathscr{E}}(\mathsf{M}^*) = q_{\mathscr{E}}(\mathsf{M})$*, equivalently* $\mathsf{M}^* \approx \mathsf{M}$*;*
(iii) $\mathscr{O}(\mathsf{M}^*) = \mathscr{O}(\mathsf{M})$*;*
(iv) $\mathscr{R}_e(\mathsf{M}^*) = \mathscr{R}_e(\mathsf{M})$*; and*
(v) M^* *has no exact latent decoder.*

Proof. Choose the Boolean extension

$$X_{\mathsf{M}^*} = X_{\mathsf{M}} \times \{0,1\}$$

and define

$$\pi_{\mathsf{M}^*}(x,b) = \pi_{\mathsf{M}}(x).$$

Define the projection

$$p : X_{\mathsf{M}^*} \to X_{\mathsf{M}}, \qquad p(x,b) = x.$$

The projection is surjective. For each $x \in X_{\mathsf{M}}$, the state $(x,0)$ maps to x. It also preserves observation because

$$\pi_{\mathsf{M}^*}(x,b) = \pi_{\mathsf{M}}(p(x,b)).$$

Thus M^* is a latent cover of M.

The cover is strict. Use the supplied witness $x_0 \in X_{\mathsf{M}}$. Then

$$(x_0,0) \neq (x_0,1)$$

but

$$p(x_0,0) = x_0 = p(x_0,1).$$

Hence the projection is noninjective.

Every latent cover is empirically equivalent to its base. Therefore

$$\mathsf{M}^* \approx \mathsf{M}$$

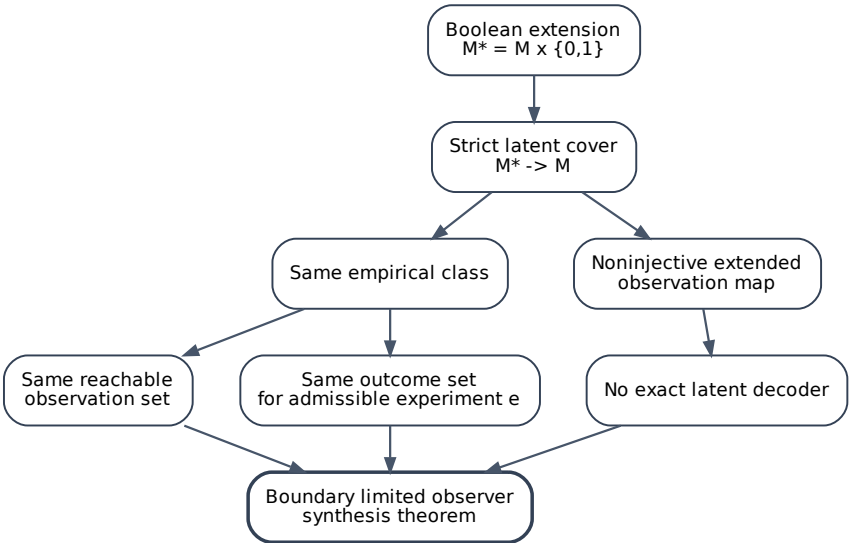


Figure 5. Dependency structure of the synthesis theorem. The result combines fiber based reconstruction limits, model outcome invariance, strict hidden extensions, and empirical quotient structure.

and

$$\mathscr{O}(\mathsf{M}^*) = \mathscr{O}(\mathsf{M}).$$

By empirical outcome invariance,

$$\mathscr{R}_e(\mathsf{M}^*) = \mathscr{R}_e(\mathsf{M}).$$

Finally, the two distinct extended states $(x_0, 0)$ and $(x_0, 1)$ have equal observations:

$$\pi_{\mathsf{M}^*}(x_0, 0) = \pi_{\mathsf{M}}(x_0) = \pi_{\mathsf{M}^*}(x_0, 1).$$

The extended observation map is noninjective. An exact latent decoder would imply injectivity. Therefore no exact latent decoder exists for M^*. □

I.1 What the theorem establishes

The theorem establishes a logical separation between empirical completeness and latent completeness. The extension agrees with the base on the entire reachable observation set. It therefore agrees under every fixed observation only experiment, including the identity experiment. The extension has distinct hidden labels and no exact decoder. If the original model is exactly decodable, their decoder statuses differ. If the original already has no decoder, both statuses are false.

The theorem also identifies the additional burden carried by a uniqueness claim. To conclude that one latent representation is uniquely determined, the argument must use assumptions not contained in reachable observation equality alone. Such assumptions may include dynamical constraints, causal interventions, regularity conditions, minimality, priors, known cardinality, or

a refined observation interface. The theorem does not decide which additional assumptions are scientifically justified. It shows why some additional condition is required.

I.2 What the theorem does not establish

The theorem does not establish that the Boolean extension is the physically realized model. It does not establish that every inaccessible distinction exists in nature. It does not establish that improved measurement can never refine the empirical class. It does not establish that approximate or probabilistic reconstruction is impossible. It establishes nonuniqueness relative to the specified observation boundary and model class.

J. Assumption checks and counterexamples

These examples establish the qualifications used in the main text. They are elementary derivations and executable regression cases, not additions to the pinned Lean formalization.

J.1 Reachable collisions in a composite map

Let $f : \{0,1\} \to \{0,1,2\}$ satisfy $f(0) = 0$, $f(1) = 1$. Let $g(0) = 0$, $g(1) = g(2) = 1$. The map g is noninjective, but $g \circ f = \mathrm{id}_{\{0,1\}}$. Its collision lies outside the restriction of g to the reachable set $f[X]$. Conversely, if distinct original states satisfy $f(x) = f(y)$, then $g(f(x)) = g(f(y))$ for every downstream function g. This proves the corrected composition statement.

J.2 Decoder status of an already noninjective base

Take $X = \{0,1\}$, $O = \{0\}$, and $\pi(x) = 0$. No exact decoder exists. Its Boolean extension also has a constant observation map and no decoder. Thus the extension theorem guarantees failure of decoding for the extension, while a change of decoder status additionally requires a decodable base.

J.3 Empty spaces and total decoders

If $O = \varnothing$, existence of $\pi : X \to O$ forces $X = \varnothing$. The empty decoder is a left inverse. If $X = \varnothing$ but $O \neq \varnothing$, no total map $d : O \to X$ exists, although π is injective. If $X \neq \varnothing$ and π is injective, fix $x_0 \in X$. Define $d(o)$ as the unique preimage for reachable o, and as x_0 otherwise. Then $d \circ \pi = \mathrm{id}_X$. These cases justify the exact-decoder condition and the nonempty assumption in the global nonidentifiability theorems.

J.4 Equal possible outputs and different probability laws

For $0 < p < 1$, the Bernoulli law has reachable outcomes $\{0,1\}$. In particular, $p = 1/10$ and $p = 9/10$ share this set but differ in $\Pr(Y = 1)$. Any extension of empirical equivalence to statistical experiments must include the probability law. Equality of reachable sets alone supplies no statement about equal frequencies or likelihoods.

J.5 Temporal reconstruction and conditioning

For $\dot{x}_1 = x_2$, $\dot{x}_2 = 0$, and $y = x_1$, the exact solution is

$$y(t) = x_1(0) + t x_2(0).$$

At time zero, all values of $x_2(0)$ share one output for a fixed $x_1(0)$. The two-sample map is

$$\begin{pmatrix} y(0) \\ y(T) \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 1 & T \end{pmatrix} \begin{pmatrix} x_1(0) \\ x_2(0) \end{pmatrix}.$$

Its determinant is T, so it is invertible when $T \neq 0$, with

$$x_1(0) = y(0), \qquad x_2(0) = \frac{y(T) - y(0)}{T}.$$

If the two output errors are bounded by η, the recovered x_2 error is at most $2\eta/|T|$. Opposite signed endpoint errors attain the bound. Invertibility alone therefore gives no uniform stability as $T \to 0$. The regression test uses rational arithmetic, so it checks the exact formulas without integration or floating point tolerances.

K. Computational methods

The executable study evaluates finite instances of the formal constructions. It does not estimate a physical parameter or fit a model to biological or physical measurements. It enumerates explicitly defined latent states, observations, experiments, predicates, and refinement maps, and then evaluates the corresponding finite equalities and inequalities.

K.1 Implementation and reproducibility

The finite study was implemented in Python. The program enumerates the declared state spaces and evaluates every observation, fiber, experiment, predicate, empirical class, and refinement relation used in the reported examples. The accompanying supplement contains the complete generated tables, selected implementation excerpts, and original simulation figures.

Table 2. Computational outputs used in the finite study.

Output group	Contents
Model summaries	Latent counts, reachable observations, fiber statistics, collision counts, injectivity, exact decoder status, and hidden multiplicity.
Experiment summaries	Possible outcome sets and equality with the injective base for four observation only experiments.
Predicate summaries	Existential success profiles and complete finite truth profiles for three accessible predicates.
Empirical quotient	Grouping of models by equality of reachable observation sets and comparison of latent properties within one class.
Refinement summaries	Interface sizes, strictness, collapsed pairs, reverse refinement, and induced quotient map properties.
Validation	Eighteen repository tests, six revision regression tests, and runner assertions covering the intended finite invariance and refinement checks.

K.2 Finite models, fibers, and collision counts

The primary finite model has latent states

$$X = \{0,1,2,3,4,5\}$$

and the identity observation map. A comparison model uses the same latent states and the rule $\pi(x) = x \bmod 3$. For every reachable observation o, the implementation enumerates the fiber

$$F_o = \pi^{-1}(o)$$

and calculates the unordered collision count

$$C(\mathsf{M}) = \sum_{o \in \mathscr{O}(\mathsf{M})} \binom{|F_o|}{2}.$$

This value is a finite count of distinct latent pairs with equal observations. It is not a distance, probability, or uncertainty measure.

K.3 Hidden multiplicity family

The Boolean extension attaches $\{\text{false}, \text{true}\}$ to each base state and ignores the added coordinate during observation. The finite family attaches

$$H_m = \{0,1,\ldots,m-1\}, \qquad m \in \{1,2,3,5,8\},$$

with observation rule

$$\pi_m(x,h) = x.$$

For the six state injective base, the finite identities are

$$|X_m| = 6m, \qquad |\mathscr{O}(\mathsf{M}_m)| = 6, \qquad |F_o| = m, \qquad C(\mathsf{M}_m) = 6\binom{m}{2} = 3m(m-1).$$

An exact decoder exists in this family only for $m = 1$.

K.4 Observation only experiments and predicates

Four experiments are functions of the accessible observation: the identity map, a threshold at three, a three category partition, and width two numerical compression. The runner evaluates each function on the complete reachable observation set of each model and records the resulting outcome set.

The predicate study uses three accessible predicates: evenness, the condition $o \geq 3$, and membership in $\{0,5\}$. The existential success profile records whether a predicate is true for at least one reachable observation. A separate truth profile records every predicate value on every reachable observation. The second object is a stronger finite diagnostic, but it is not definitionally identical to the formal existential predicate profile.

K.5 Finite empirical quotient and interface refinement

The finite empirical quotient groups models by exact equality of reachable observation sets. The grouping rule does not compare latent cardinality, fiber size, decoder status, or hidden factor metadata.

The refinement study uses one eight state world and four interfaces: identity, paired buckets, binary regions, and a relabeled identity. Each proposed refinement is checked on every world state. The implementation then records strictness, collapsed state pairs, reverse refinement, equality of observational resolution, and surjectivity, injectivity, and bijectivity of the induced finite quotient map.

K.6 Automated validation

The reference implementation includes eighteen automated tests. This revision adds six tests of assumption boundaries, supplied in the source package. They cover injective and noninjective base models, Boolean and finite hidden extensions, equality of experiment outcome sets, predicate profiles, empirical quotient grouping, observation fibers, refinement preservation, strict collapse, quotient map surjectivity and noninjectivity, reverse refinement failure, relabeling equivalence, composition, and invalid coarsening rejection. The runner also contains explicit assertions that stop execution if any intended finite relation fails.

K.7 Formal and computational verification boundary

The pinned repository records successful Lean checking against mathlib. This revision inspects the encoded statements without claiming a new local Lean build. The finite examples are rerun by exhaustive Python enumeration. These procedures serve different roles. The proof checker verifies derivability within the encoded formal system. The simulations verify the stated equalities and counts for the enumerated finite models. Neither procedure establishes that the formal encoding is the unique description of a physical system or that the simulated interface is biologically or physically complete.

This separation follows from the interface framework. Every computation requires a representation of physical or conceptual states, an allowed evolution, and a readout (Horsman et al., 2014; Horsman, 2015). The verified theorem is exact relative to its definitions. A proposed application must additionally justify the map from the application domain into those definitions and quantify approximation, noise, and omitted variables.

L. Computational results

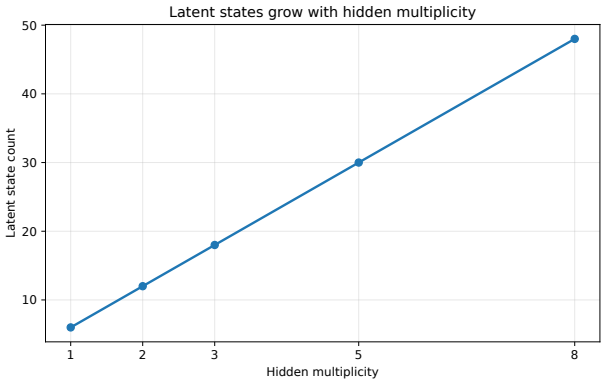
The finite calculations satisfy the stated identities and invariance checks. The tables report exact enumerations, and the figures display the corresponding results for the multiplicity and refinement families.

L.1 Base models and Boolean extension

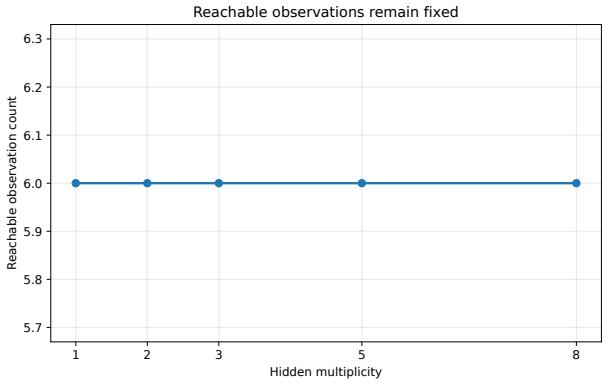
The injective base contains six latent states and six reachable observations. Every fiber has size one, the observation map is injective, and an exact decoder exists. The noninjective comparison contains six latent states and three reachable observations. Each reachable fiber has size two, the model contains three unordered latent collisions, and no exact decoder exists. The Boolean extension

Table 3. Base and Boolean extension calculations.

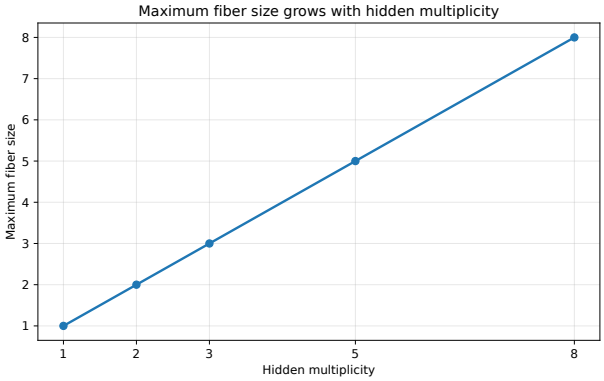
Model	Latent	Reachable	Max. fiber	Collisions	Injective	Exact decoder
Injective base	6	6	1	0	True	True
Noninjective base	6	3	2	3	False	False
Boolean extension	12	6	2	6	False	False



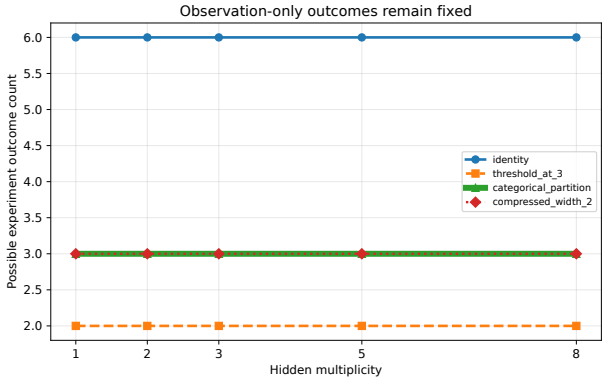
(a) Latent state count.



(b) Reachable observation count.



(c) Maximum fiber size.



(d) Observation only outcome counts.

Figure 6. Hidden multiplicity and observation only experiment results. The latent representation and observation fibers grow while the reachable observation set and each experiment's possible outcome count remain fixed.

contains twelve latent states and the same six reachable observations as the injective base. Its maximum fiber size is two, it contains six collision pairs, and no exact decoder exists.

L.2 Finite hidden multiplicity

The latent state count increases linearly with hidden multiplicity. The reachable observation count remains fixed at six, and the maximum fiber size equals the hidden multiplicity. The four panels show the latent-state, reachable-observation, fiber-size, and experiment-outcome consequences of the same multiplicity construction.

Every row is marked empirically equivalent to the base. Latent cardinality, fiber size, collision count, injectivity, and exact decoder status change while the reachable observation set remains fixed.

Table 4. Complete finite multiplicity summary.

m	Latent	Reachable	Avg. fiber	Max. fiber	Collisions	Exact decoder
1	6	6	1.00	1	0	True
2	12	6	2.00	2	6	False
3	18	6	3.00	3	18	False
5	30	6	5.00	5	60	False
8	48	6	8.00	8	168	False

Table 5. Observation only outcome sets.

Experiment	Possible outcomes	Count	All listed models match base
Identity	$\{0,1,2,3,4,5\}$	6	True
Threshold at three	$\{\text{false}, \text{true}\}$	2	True
Categorical partition	$\{\text{high}, \text{low}, \text{middle}\}$	3	True
Compression width two	$\{0,1,2\}$	3	True

Table 6. Base predicate profiles and comparison counts.

Predicate	Base truth table on $0,\ldots,5$	Models compared	Matches
Even	T, F, T, F, T, F	7	7
At least three	F, F, F, T, T, T	7	7
Endpoint $\{0,5\}$	T, F, F, F, F, T	7	7

L.3 Observation only experiments and predicate profiles

The identity experiment has six possible outcomes. The threshold experiment has two. The categorical partition and width two compression experiments each have three. All twenty eight model and experiment comparisons match the base outcome set for the corresponding experiment.

Three predicates are evaluated for seven model representations. All twenty one existential comparisons match the base success profile. All twenty one complete truth profile comparisons also match the base.

L.4 Finite empirical quotient and varying latent properties

The finite grouping procedure produces two empirical classes. The first contains the injective base, Boolean extension, and finite multiplicity extensions. Its reachable set is $\{0,1,2,3,4,5\}$, while its representatives have latent cardinalities $6, 12, 18, 30, 48$. The second contains the noninjective comparison model with reachable set $\{0,1,2\}$.

Four latent properties vary between the injective base and its Boolean extension although the two models occupy the same finite empirical class.

These finite tables are witnesses inside the framework. They do not replace the general proof.

L.5 Interface refinement and quotient maps

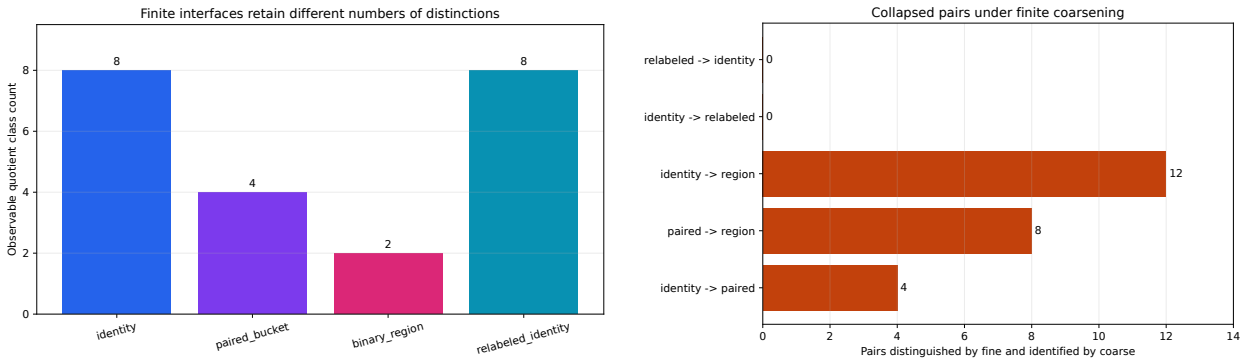
The identity and relabeled identity interfaces retain eight observable classes. Paired buckets retain four. Binary regions retain two. The strict refinements collapse four, eight, and twelve world state

Table 7. Finite empirical quotient.

Class	Reachable observation set	Latent cardinalities	Representative types
1	$\{0,1,2,3,4,5\}$	6, 12, 18, 30, 48	Base, Boolean, and finite multiplicity extensions
2	$\{0,1,2\}$	6	Noninjective comparison model

Table 8. Properties that vary inside one finite empirical class.

Property	Base	Boolean extension	Varies
Latent cardinality	6	12	True
Maximum observation fiber size	1	2	True
Existence of exact latent decoder	True	False	True
Presence of Boolean hidden factor	False	True	True



(a) Observable quotient size.

(b) Collapsed state pairs.

Figure 7. Interface refinement results. Strict coarsening reduces the retained quotient classes and identifies world state pairs. Relabeling preserves observational resolution.

pairs. Their quotient maps are surjective and noninjective, and no reverse refinement exists. The relabeling maps collapse no pairs and induce bijections.

The complete quotient-map enumeration contains thirty six class-map entries, and the strict-refinement enumeration contains twenty four collapsed-pair records. Both complete tables are reproduced in the supplement.

Table 9. Finite interface and refinement summary.

Refinement	Fine Q	Coarse Q	Collapsed	Strict	Surjective	Injective
Identity to paired	8	4	4	True	True	False
Paired to region	4	2	8	True	True	False
Identity to region	8	2	12	True	True	False
Identity to relabeled	8	8	0	False	True	True
Relabeled to identity	8	8	0	False	True	True

L.6 Validation result

All eighteen repository tests and all six revision regression tests pass. All twelve regenerated CSV outputs match the archived files byte for byte. The executable runner also verifies Boolean extension, multiplicity, experiment, predicate, empirical quotient, and interface refinement checks before writing the output files.

M. Operational architecture

M.1 Process, feedback, and perspectival description

The architecture is intentionally process-based. Whitehead's process philosophy is a historical precedent for treating events, transitions, and relations as primary descriptive units rather than reducing the analysis to static substances (Whitehead, 1978). Wiener's cybernetics supplies the explicit language of feedback, communication, and control (Wiener, 1948). Von Neumann's analyses of measurement and self-reproducing automata show why an observer, apparatus, description, and update rule cannot always be treated as independent endpoints (von Neumann, 1955, 1966). Leibnizian perspectivism provides an earlier comparison for the idea that one world may admit multiple internally coherent representations (Leibniz, 1989). None of these sources is identical to the map developed below. They explain why the complete object is a recursive transformation system rather than a single passive observation.

Bohm's distinction between implicate and explicate order and the Bohm–Hiley account of undivided quantum process are retained as comparative physical programs, not as assumptions of the present analysis (Bohm, 1980; Bohm and Hiley, 1993). The present theory requires only declared maps and equivalence relations. It does not infer an implicate order, a quantum ontology, or a universal process metaphysics from noninjectivity.

M.2 The complete recursive scientific cycle

The formal theory is a local core inside a larger scientific cycle. Let $S(t)$ denote a physical or modeled state at time t. Let A_i denote an apparatus, intervention, or sampling procedure. Let E_i denote a finite encoding or reconstruction step. Let O_i denote the representation accessible to an observer. Let L_i denote a reportable model, estimate, or claim. Let P_i denote a prediction, intervention, or next experimental choice. One cycle has the form

$$S(t) \longrightarrow A_i \longrightarrow E_i \longrightarrow O_i \longrightarrow L_i \longrightarrow P_i \longrightarrow S(t+\Delta t).$$

The final state enters the next cycle through new measurement. The complete process is recursive. This perception-action-return structure is compatible with predictive-processing and active-inference accounts in which observations update a generative model and actions change subsequent observations (Rao and Ballard, 1999; Friston, 2010; Parr et al., 2022; Friston et al., 2025).

A physical state is not identical to an apparatus output. An apparatus output is not identical to a reconstructed feature. A reconstructed feature is not identical to an internal representation or report. A report is not identical to the next state of the system. Each arrow requires an explicit map.

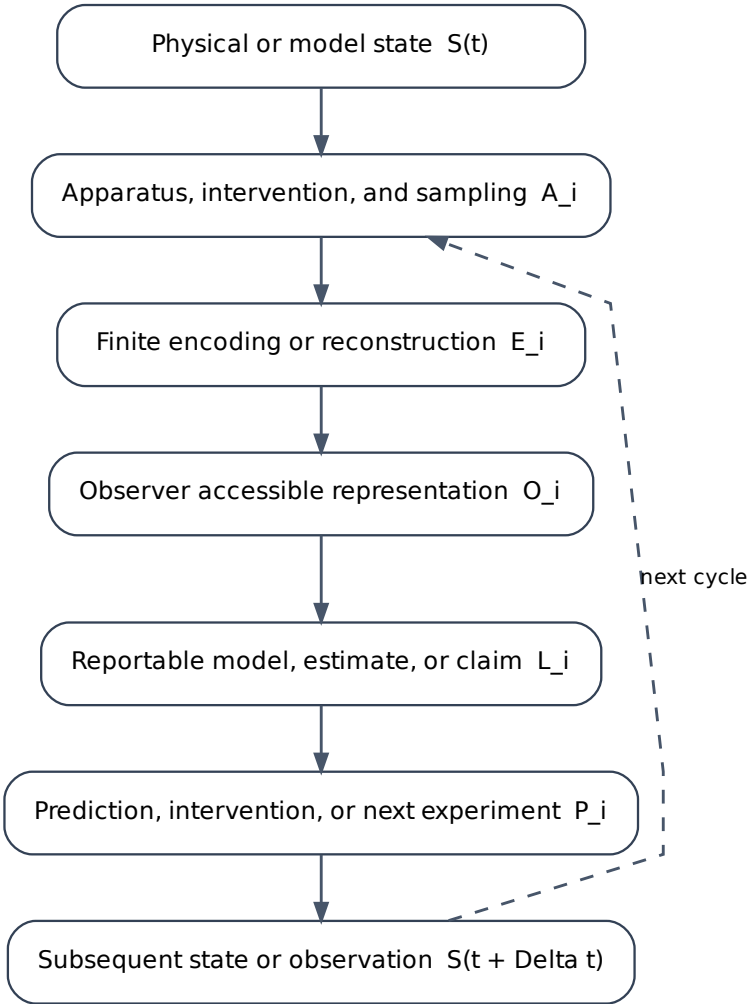


Figure 8. Top-down scientific architecture. The map studied in this paper is the reduced local core of a longer recursive cycle. The dashed return marks prediction, intervention, and subsequent observation. The diagram is conceptual rather than a simulation output.

M.3 A distributed family of boundaries

The complete route is better represented as a composition:

$$X_{\text{phys}} \xrightarrow{\pi_{\text{app}}} O_{\text{app}} \xrightarrow{\pi_{\text{bio}}} O_{\text{bio}} \xrightarrow{\pi_{\text{rep}}} O_{\text{rep}} \xrightarrow{e} R \xrightarrow{a} X'_{\text{phys}}.$$

The maps may correspond to physical coupling, detector response, sampling, reconstruction, biological sensing, attention, memory, language, or experimental decision. The aggregate observation map is

$$\pi = \pi_{\text{rep}} \circ \pi_{\text{bio}} \circ \pi_{\text{app}}.$$

If two distinct original source states reach the same intermediate value, every subsequent deterministic map preserves that equality. The composite is then noninjective. Noninjectivity of a stage outside the range reached by preceding stages is insufficient; a counterexample is given in Appendix J. Refining a later report cannot recover a distinction that was already collapsed upstream unless new information enters through another route.

The stages should eventually be modeled separately rather than compressed into one aggregate map. Apparatus response, sampling, biological transduction, internal representation, report, and action may each preserve different relations and collapse different distinctions. Stage-specific models would show where a loss first occurs and whether later measurements introduce genuinely new information or only re-encode what has already passed through an earlier boundary.

M.4 Computation as an interface-bound physical process

A computation implemented in the world is not identical to its abstract description. It requires a physical encoding, a physical evolution, and a readout:

$$S_{\text{phys}} \xrightarrow{\eta} C \xrightarrow{U} C' \xrightarrow{\rho} R.$$

Here η assigns an abstract computational state to a physical condition, U is the declared abstract evolution, and ρ converts a final physical or encoded state into a reportable result. Frameworks for physical computation make this representation relation explicit and distinguish the abstract calculation from the device that realizes it (Horsman et al., 2014; Horsman, 2015). When the computation is embedded inside the system being studied, both η and ρ are themselves observer interfaces.

The distinction matters because different physical computers do not expose the same state space, elementary operations, error model, or readout. A digital computer represents finite symbolic states through engineered voltage ranges. An analog computer uses calibrated continuous physical variables to approximate a differential relation (Shannon, 1941). A quantum computer manipulates quantum states but ultimately requires a controlled physical implementation and a classical measurement interface (DiVincenzo, 2000; Aharonov et al., 2022). A brain–computer interface converts selected electrophysiological activity into device commands and requires adaptation of both the user and decoder (Wolpaw et al., 2002; Willett et al., 2023). Human reasoning is mediated by sensory transduction, neural dynamics, memory, language, and report. Equality of an output across these systems does not imply equality of their internal logic, physical representation, or relation to the world.

A sensor suite gives a concrete example of a distributed boundary. No individual sensor measures the complete state of a vehicle or environment. Each channel returns a partial representation with its own bandwidth, geometry, calibration, noise, and blind regions. Multisensor fusion combines those reports into an estimated state or decision, but the fusion process does not remove the interface. The result remains conditional on sensor coverage, synchronization, uncertainty models, and the rule used to combine observations (Hall and Llinas, 1997; Durrant-Whyte and Henderson, 2008).

The familiar joke that a missile "knows where it is because it knows where it is not" is not a technical account of guidance. Its useful core is the role of discrepancy. A sensor suite used to guide or track a missile, a reentry vehicle carrying a warhead, or a spacecraft compares an evolving state estimate with incoming measurements and reference constraints. The residual identifies

Table 10. Examples of physically distinct interfaces. The rows are not claimed to be mathematically equivalent. They show why a result must be interpreted relative to the encoding, accessible state, readout, and failure conditions of the particular interface.

Interface	Accessible state and readout	Principal boundary
Digital computer	Finite symbolic words encoded by engineered voltage ranges and returned through a discrete output channel.	Word length, architecture, numerical precision, software semantics, and the mapping between physical voltage and symbol.
Analog computer	Continuous voltages, currents, rotations, or other calibrated quantities interpreted as variables in a differential model.	Noise, drift, dynamic range, calibration, and the approximation relating physical trajectory to mathematical solution.
Quantum computer	Prepared quantum states, controlled transformations, and measurement outcomes converted into classical records.	State preparation, decoherence, permitted measurements, sampling, error correction, and the quantum–classical readout boundary.
Brain–computer interface	Selected neural signals translated into commands, text, movement, or another device output.	Electrode placement, signal preprocessing, decoder training, adaptation, and the distinction between intended state and decodable signal.
Human sensory and cognitive interface	Receptor signals integrated through neural dynamics, memory, attention, language, and report.	Biological bandwidth, selective attention, prior structure, state dependence, and reportability.
Multisensor suite and fused state estimate	Heterogeneous measurements combined with a dynamical model or reference data into an estimated state, residual, confidence value, or track.	Coverage, calibration, synchronization, bias, correlated error, fusion rule, model mismatch, and unobserved state components.
Inertial guidance and state estimation	Accelerometer and gyroscope signals combined with a dynamical model to maintain an estimated position, velocity, attitude, and error state.	Sensor bias, drift, model mismatch, initialization, reference updates, and uncertainty in the estimated state.
Particle detector and analysis pipeline	Detector interactions reconstructed into tracks, energies, event classes, likelihoods, and parameter limits.	Acceptance, efficiency, resolution, trigger selection, reconstruction assumptions, background model, and statistical decision rule.

disagreement between prediction and accessible evidence; it does not reveal an unrestricted external ground truth. The system therefore acts from a bounded coordinate model and quantified uncertainty rather than direct possession of its complete physical state (Kalman, 1963; Hall and Llinas, 1997; Titterton and Weston, 2004). This is an interface example, not an operational description of weapon design or use.

Particle physics also requires several meanings of a "forbidden zone" to be kept separate. A region may be kinematically forbidden by a specified theory, outside detector acceptance, unob-

Table 11. Taxonomy of outcomes across an observation or representation map.

Outcome	Structural meaning	Mathematical or empirical question
Preservation	A relation, value, or invariant survives the map.	Which quantities factor through the interface, and are they complete for the target task?
Re-encoding	Structure is expressed in a different representation.	Is the transformation invertible, approximately invertible, or only relation-preserving?
Compression	The accessible representation contains fewer distinguishable classes than the source.	Which fibers are enlarged, and what refinement would split them?
Dispersion	Correlations or organized structure spread across degrees of freedom, time, or scale.	Which observables still carry the relation, and at what scale or resolution?
Collapse	Distinct source states map to one accessible state.	Is the map noninjective, and what latent properties vary inside a fiber?
Inaccessibility	A structure may remain in the source but lies outside the declared observation channel or threshold.	Does another interface represent it, or is its existence itself underdetermined?
Physical destruction	The source structure no longer exists under a domain-specific criterion.	What physical law, measurement, or dynamical model establishes destruction rather than nonrepresentation?

servable because of a blind or dead region, or statistically excluded by data under a model and confidence procedure. Experimental exclusion regions are therefore conditional on the detector, event reconstruction, background model, parameterization, and statistical rule (Cowan et al., 2011; Particle Data Group, 2026). Exclusion is a strong scientific result, but it is not the same statement as absolute nonexistence independent of the declared interface and model.

Formal exactness and physical approximation must likewise be separated. A proof can be exact relative to its encoded definitions and inference rules, while every physical implementation of the proof checker, sensor, or simulation remains dependent on finite hardware, representation, and readout. Mechanical verification establishes that the encoded conclusion follows from the encoded premises. It does not establish that the encoding is the only possible formalization of human reasoning or that a formal state is identical to the brain's reality interface.

M.5 Preservation, transformation, compression, dispersion, and collapse

The phrase *information loss* does not identify what a map has done. The following taxonomy is used throughout the interpretation and future-work sections.

A relation can be preserved while its carrier changes. A signal can be transformed without being destroyed. A source distinction can be real but inaccessible to one interface. Conversely, a mathematically possible hidden distinction need not be physically real. The taxonomy prevents *unobserved*, *compressed*, *dispersed*, and *destroyed* from being used as synonyms.

M.6 External analyst and embedded observer

Two observational positions must be distinguished. An external analyst can define a model, inspect its source space, and compare it with the observation map. An embedded observer is itself represented within the physical and informational process it is attempting to model. Its self-model, measurement tools, and inference rules are generated and updated within the same larger dynamics.

The present formalization uses the external analyst's mathematical viewpoint to prove facts about a declared interface. It does not yet formalize a system whose model of itself modifies the system that performs the modeling. That embedded problem is closer to second-order cybernetics, autopoiesis, active inference, and recent work on self-referential living systems (von Foerster, 1981; Maturana and Varela, 1980; Friston, 2010; Abramsky et al., 2025). It is identified in Appendix O as the next major formal extension.

M.7 Reduction to the formal core

The current theorem sequence isolates

$$X \xrightarrow{\pi} O \xrightarrow{e} R,$$

where X is the source or latent state space, O is the accessible observation space, and $e : O \to R$ is an admissible experiment. The reduced map may represent one boundary or the composition of many. It is not a replacement for the complete cycle. It is the smallest local object on which the reconstruction question can be stated and proved.

M.8 Three levels of description

The mathematical core keeps three object levels separate. A *source state* is an element of X. An *accessible observation* is an element of O. An *experimental result* is an element of R. Two source states can produce one observation. Two models can produce one reachable observation set. Several observations can produce one result. Each equality is relative to the map that defines it.

M.9 Local noninvertibility

For $o \in O$, the fiber is

$$\pi^{-1}(o) = \{x \in X : \pi(x) = o\}.$$

When this set contains multiple states, the observation identifies the fiber but not a unique member.

The accessible value contains no internal tag selecting its source member. A reconstruction claim must therefore use an injective interface, an independently justified restriction on the source class, or additional information supplied from outside the current observation value.

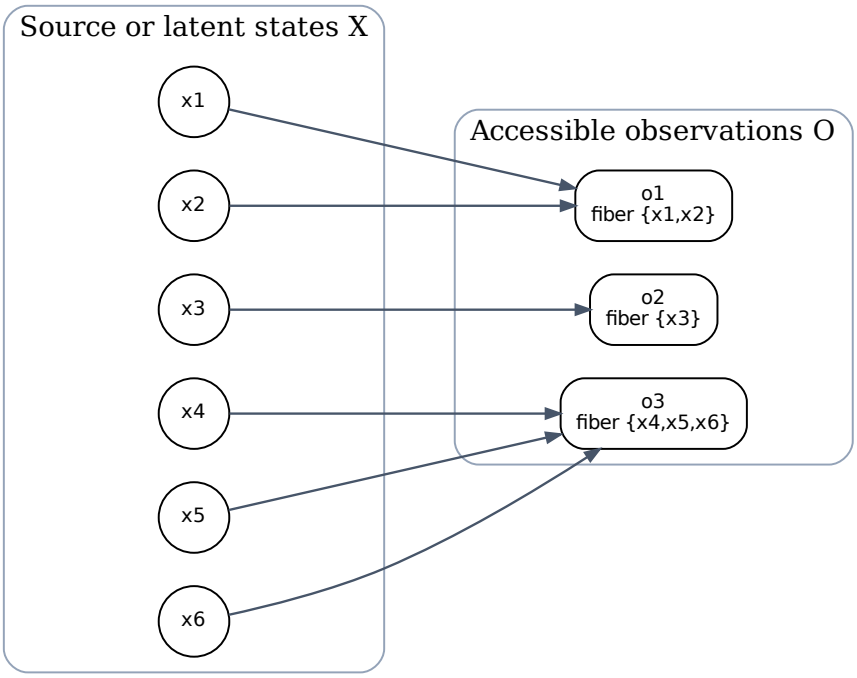


Figure 9. Observation fibers. States in one fiber produce the same accessible value. Fiber membership is relative to the specified map. A refined interface may split a fiber that a coarse interface leaves merged.

N. Related work

The paper sits at the intersection of several traditions. The comparison must remain exact: neighboring work motivates questions and supplies established results, but it does not become evidence for the new theorem merely because it uses similar words.

N.1 Appearance, causation, and the conditions of representation

Plato's cave is an early model of mediated appearance: the experienced display is real as experience without exhausting the structure that produces it (Plato, 1992). Aristotle's analysis of cause warns against treating one explanatory relation as a complete account of an object (Aristotle, 1984). Husserl's phenomenology examines how objects of science are constituted relative to intentional structures and a prior lifeworld (Husserl, 1970). The present analysis does not adopt Platonic forms, Aristotelian metaphysics, or transcendental phenomenology. These sources clarify why the paper separates source state, accessible representation, explanatory level, and ontology.

N.2 Leibnizian perspective, sufficient reason, and indiscernibility

Leibniz is a central precursor. In the *Monadology* and related essays, individual monads express one world from distinct points of view; his principle of sufficient reason asks what grounds the selection of one account rather than another; and the identity of indiscernibles asks whether genuinely distinct entities can share all properties (Leibniz, 1989; Leibniz and Clarke, 2000). The present interface

question reverses and restricts the last question. It asks whether states can remain distinct while sharing every property accessible through a declared interface. The answer is yes whenever the interface has a nontrivial fiber. This is not a refutation of Leibniz because "all accessible predicates" is weaker than "all properties." It is a formal statement of why the distinction matters.

Leibniz's project of a universal characteristic and a calculus of reasoning is also relevant to machine-checked formalization. Lean is not a realization of that full project, but the aspiration to separate symbolic derivation from informal persuasion belongs to the same historical line.

N.3 Process, relation, and hidden order

Whitehead treats reality as process and relational becoming rather than a collection of independently fixed substances (Whitehead, 1978). Bohm develops wholeness and implicate order, and Bohm with Hiley presents an ontological interpretation of quantum theory in which accessible phenomena are related to a deeper undivided process (Bohm, 1980; Bohm and Hiley, 1993). These sources motivate attention to transformations and preserved relations. The present analysis does not assume process metaphysics or Bohmian quantum theory. Its verified claims hold for ordinary sets and functions.

N.4 Diagonalization, self-reference, and semantic closure

Cantor's diagonal method established a general technique for producing an object outside a proposed enumeration (Cantor, 1891). Russell exposed a contradiction in unrestricted self-membership (Russell, 1908). Gödel proved incompleteness for sufficiently expressive formal theories (Gödel, 1931). Tarski proved limits on internally defined truth predicates (Tarski, 1936). Lawvere expressed a diagonal fixed-point schema categorically (Lawvere, 1969). Hofstadter explored recursive self-description and strange loops as an explanatory pattern spanning formal systems and minds (Hofstadter, 1979, 2007). Penrose used Gödelian arguments to motivate stronger claims about mind and computation (Penrose, 1989, 1994); those stronger conclusions remain disputed and are not assumed here.

The present analysis does not derive any of these theorems from an observation map, and they do not prove that a physical observer must have a noninjective interface. Their relevance is structural: they distinguish local formal limitations from unrestricted claims about reality and show why a general embedded-boundary invariant requires more than analogy.

N.5 Cybernetics, self-production, and observing systems

Wiener placed communication, control, and feedback in one mathematical program (Wiener, 1948). Ashby treated regulation and requisite variety as relations between disturbances and available responses (Ashby, 1956). Second-order cybernetics included the observer within the act of observing (von Foerster, 1981). Autopoietic theory emphasized organizational closure and the difference between a system's own organization and an external description (Maturana and Varela, 1980). Rosen's relational biology asked how living organization differs from a merely external mechanism (Rosen, 1991). Von Neumann's work on measurement and self-reproducing automata supplies an additional bridge between description, observer participation, and self-maintaining organization (von Neumann, 1955, 1966).

These traditions motivate the embedded-observer problem. They do not provide the state-level observable quotient, model-level empirical quotient, and hidden-extension theorem sequence developed here.

N.6 Observability, identifiability, and inverse problems

Kalman observability concerns recovery of a dynamical state from an output trajectory in a system with specified dynamics (Kalman, 1963). Hermann and Krener extended observability analysis to nonlinear systems (Hermann and Krener, 1977). Structural identifiability asks whether distinct parameter values or model structures produce the same idealized observations (Bellman and Åström, 1970; Rothenberg, 1971; Walter and Pronzato, 1997). Applied work distinguishes structural nonidentifiability from practical uncertainty due to noise or finite data (Calderón, 1980; Raue et al., 2009; Chis et al., 2011; Villaverde et al., 2016; Meshkat et al., 2009; Guillaume et al., 2019).

Hadamard's well-posedness conditions and later regularization theory distinguish nonexistence, nonuniqueness, and instability (Hadamard, 1923; Tikhonov, 1963; Engl et al., 1996). The present result isolates nonuniqueness at the interface level. Regularization may select a representative, but selection is not proof that the observation uniquely determines it.

N.7 Statistical experiments, automata, and causal equivalence

Blackwell comparison orders statistical experiments by informativeness for decision problems (Blackwell, 1951, 1953; Torgersen, 1991). Moore experiments and bisimulation compare states or systems by designated external behavior (Moore, 1956; Park, 1981; Milner, 1989; Larsen and Skou, 1991; Rutten, 2000). Causal graphical models admit observational Markov-equivalence classes that can be refined by intervention or additional assumptions (Verma and Pearl, 1990; Chickering, 1995; Andersson et al., 1997; Pearl, 2009; Hauser and Bühlmann, 2012).

The present formalization uses a deterministic reachable-set relation rather than probability kernels, utilities, transition traces, or causal independences. The common lesson is that equivalence is relative to a specified family of observations or tests.

N.8 Latent representation learning and abstraction

Finite mixtures, multilinear decompositions, and modern latent-variable models require explicit uniqueness conditions (Teicher, 1963; Kruskal, 1977; Allman et al., 2009; Anandkumar et al., 2014). Unsupervised disentanglement is not identifiable in general without inductive bias (Locatello et al., 2019), while auxiliary variables, multi-view structure, or stronger restrictions can restore identifiability (Hyvärinen et al., 2019; Khemakhem et al., 2020; Gresele et al., 2020; Xi and Bloem-Reddy, 2023). Tenenbaum and colleagues frame learning as the growth of structured abstractions rather than direct copying of sensory data (Tenenbaum et al., 2011). The hidden-factor construction supplies a deterministic witness for the broad warning that latent semantics do not follow from observational agreement alone.

N.9 Information, coarse-graining, constraint, and scale

Information theory separates a source from its transmitted representation and studies what processing preserves or reduces (Shannon, 1948a,b; Kullback and Leibler, 1951; Cover and Thomas, 2006). Nonequilibrium self-organization, complex adaptive systems, and constraint-based emergence provide different accounts of how stable organization can persist across scales (Nicolis and Prigogine, 1977; Kauffman, 1993; Deacon, 2011; Mitchell, 2009). Recent scale-free active-inference work constructs hierarchical generative models in which coarse-grained states and paths are recursively composed across levels (Friston et al., 2025).

These approaches are directly relevant to preservation, re-encoding, compression, and dispersion. The present formalization defines none of their thermodynamic, probabilistic, or dynamical quantities. A future connection must specify the map and prove which invariants descend through it.

N.10 Axiomatics, law, and invariance

Hilbert's axiomatic geometry demonstrates how a mathematical structure can be characterized by relations that admit different realizations (Hilbert, 1902). Newton's laws and Einstein's special relativity provide distinct models of physical law and invariance across admissible descriptions (Newton, 1999; Einstein, 1905). Feynman's lectures on physical law emphasize that compact mathematical structure is not a substitute for empirical testing (Feynman, 1965). These examples motivate the search for invariants across interfaces while guarding against the inference that every formal invariant is physically realized.

N.11 Observer participation in physics and computation

Wheeler's "law without law" program places observation and participatory description near the foundations of physics (Wheeler, 1983). Constructor theory reframes physical laws in terms of possible and impossible transformations, including information-bearing transformations (Deutsch, 2013; Deutsch and Marletto, 2015). Tegmark asks whether consciousness can be characterized as a state of matter with specified information-processing properties (Tegmark, 2015). Wolfram's computational program emphasizes computational irreducibility and develops an explicit observer theory in which bounded observers aggregate computational histories (Wolfram, 2002, 2023; Rickles et al., 2023).

These programs differ sharply in ontology and mathematics. They are included because each makes the observer, transformation, or coarse-grained description an explicit part of the explanatory object. The present formalization does not infer physical law from computation and does not treat observer equivalence as evidence for any one foundational physics.

N.12 Physical computation and heterogeneous readout

Shannon's theory of the differential analyzer treats continuous physical evolution as an analog calculation (Shannon, 1941). Horsman and colleagues distinguish the physical device from the abstract computation by an explicit representation relation (Horsman et al., 2014; Horsman, 2015). Quantum computation adds a further separation among state preparation, controlled evolution, quantum measurement, and classical output (DiVincenzo, 2000; Aharonov et al., 2022). Brain–computer interfaces translate selected neural activity into device commands and require adaptation across the biological and computational sides of the interface (Wolpaw et al., 2002; Willett et al., 2023).

These cases support a general caution: two devices may compute the same abstract function while differing in substrate, accessible state, noise, resources, and readout. Conversely, two interfaces may produce the same report without implementing the same process. The theorem concerns the declared observation relation and therefore applies only after the encoding and readout maps have been stated.

N.13 Consciousness theories, interfaces, and current empirical tests

Chalmers distinguishes functional explanation from the explanatory problem of subjective experience (Chalmers, 1995). Integrated information theory, global workspace approaches, attention

schema theory, resonant neural dynamics, predictive processing, and active inference offer different accounts of access, integration, report, or experience (Tononi, 2008; Koch, 2019; Graziano, 2013; Grossberg, 2021; Friston, 2010; Parr et al., 2022). Jaynes provides a historically influential, controversial account of changes in self-modeling and source attribution (Jaynes, 1976). Determinist and compatibilist accounts of agency frame a related level-of-description dispute: Sapolsky emphasizes the causal history of biological action, whereas List, Dennett, and Frankfurt defend forms of agency that do not require exemption from physical causation (Sapolsky, 2023; List, 2019; Dennett, 2003; Frankfurt, 1971). Hoffman and Prakash propose interface theory and conscious-agent models in which perceptual representations do not transparently reproduce objective structure (Hoffman and Prakash, 2014; Prentner and Hoffman, 2024).

Recent debate has emphasized explicit theory comparison and the danger of pseudo-operationalization (Gomez-Marin and Seth, 2025; Tononi et al., 2025). The 2025 COGITATE adversarial collaboration preregistered divergent predictions of integrated information theory and global neuronal workspace theory and found results that challenged central predictions of both (Cogitate Consortium et al., 2025). Laukkonen, Friston, and Chandaria propose a recursively shared world model and precision control as conditions for consciousness (Laukkonen et al., 2025). Whyte and colleagues formulate a minimal testable consciousness theory implicit in active inference (Whyte et al., 2026). Robinson and colleagues report an adversarial test design for an active-inference account of conscious content (Robinson et al., 2025).

The present theorem does not prove a theory of consciousness. It supplies a reconstruction and equivalence language in which claims about unified state, reportability, hidden mechanism, and competing implementations can be separated.

N.14 Artificial systems and machine consciousness

Chalmers analyzes obstacles and possible future routes to consciousness in large language models (Chalmers, 2023). Butlin and colleagues derive computational indicator properties from several neuroscientific theories and use them to structure assessment of artificial systems (Butlin et al., 2023). These works are relevant to the embedded-interface question because an artificial system may expose reports, hidden states, and self-monitoring channels that are neither identical nor mutually reconstructable. The present theorem does not decide whether any artificial system is conscious. It states what cannot be inferred from equality at a selected output interface alone.

N.15 Recent generalized observers and self-referential living systems

Elshatlawy and colleagues propose a cross-disciplinary formal theory of observers with operational observer equivalences and complexity measures (Elshatlawy et al., 2025). Abramsky and colleagues identify time, self-reference, and self-modification as open formal problems for living systems and discuss domain theory, coalgebra, and self-modifying computation as possible tools (Abramsky et al., 2025). Both are developing programs rather than settled results.

Their relevance is direct. The next formal extension must distinguish an external analyst from an embedded observer and model a self-interface whose outputs feed back into the system that generates them.

N.16 Neural manifolds and reportable structure

Toroidal organization in grid-cell population activity supplies a concrete example of structure recovered through a long measurement and reconstruction chain rather than observed without mediation

(Gardner et al., 2022). Related work has recovered toroidal organization in one-dimensional behavior and investigated how oscillatory dynamics contribute to the observed topology (Hermansen et al., 2024; di Sarra et al., 2025a). In each case, electrodes, sampling, spike sorting, behavioral variables, preprocessing, dimensionality reduction, and topological analysis participate in the accessible representation.

The oscillation study has a published correction replacing duplicated panels in its synthetic torus calibration figure (di Sarra et al., 2025b). This calibration is distinct from both its neural recordings and the finite set calculations presented here.

The relevance is methodological. A neural manifold can be a stable and informative empirical structure without being identical to the complete neural or experiential state that produced it. Future analysis should declare each map in the data-to-representation chain, test whether the recovered topology survives justified changes in preprocessing and sampling, and determine which latent alternatives remain observationally equivalent.

N.17 Ultraweak photon emission and the light hypothesis

Ultraweak photon emission is a measurable phenomenon associated with oxidative and metabolic processes in living systems. Recent work reports spontaneous emission from DNA under controlled conditions and measurable differences between living and dead mice, with stress-related changes in plants (Pietruszka and Marzec, 2024; Salari et al., 2025). Reviews summarize detection methods and proposed biological applications (Mould et al., 2024).

These results do not establish that light is a universal informational standard of life, that biological information is conserved after death in an observer-independent sense, or that photon emission carries the complete organization of an organism. They identify a concrete observable that can enter a future scale-transition study without turning the motivating hypothesis into a present conclusion.

N.18 Formal verification and encoded proof

Lean provides a dependent-type-theory environment in which definitions and proofs can be checked mechanically (de Moura et al., 2015). Mathlib supplies reusable formal mathematics (The mathlib Community, 2020), and projects such as Flyspeck demonstrate the role of mechanical checking in substantial proofs (Hales et al., 2017). The historical aspiration to mechanize valid reasoning also recalls Leibniz's universal-characteristic program (Leibniz, 1989).

Mechanical checking certifies a relation between encoded premises, inference rules, and conclusions. It does not make the proof checker an unrestricted observer of physical reality, and it does not imply that classical symbolic logic, analog dynamics, quantum measurement, neural computation, and human reasoning share one representation interface. Physical-computation frameworks explicitly separate an abstract computational description from the device-level encoding and readout that realize it (Horsman et al., 2014; Horsman, 2015). The distinction is therefore the same one used throughout the paper: correctness within an interface is not identity with every latent process that may implement, interpret, or motivate it.

N.19 Distinct contribution

The mathematical contribution is the combination of two quotient levels and an explicit counter-model construction. At the state level, an interface creates fibers and an observable quotient. At the model level, equality of reachable sets creates an empirical quotient. Hidden extensions move within

one empirical class while changing latent structure and exact-decoder status. Interface refinement orders boundaries by the distinctions they preserve. The synthesis theorem packages these relations into one verified statement. The wider genealogy explains why that bounded result matters, but no historical or contemporary theory is treated as having proved it in advance.

O. Broader applications and research hypotheses

This appendix retains the broader research program. Its hypotheses are not consequences of the reconstruction theorem. The formal framework is exact, deterministic, static, and intentionally bounded. The limitations below define both the assumptions required for broader applications and the research program that follows.

O.1 Set-based empirical equivalence and exact equality

Empirical equivalence is equality of reachable observation sets. Real measurements include probabilities, frequencies, noise, finite precision, and time dependence. Two models can have the same support and different probability laws. A stochastic extension should assign measures or kernels to latent and observation spaces. Approximate equivalence should replace exact fibers by metric neighborhoods, couplings, or reconstruction-error bounds.

O.2 Static interfaces and unrestricted hidden factors

A dynamical trajectory can distinguish states that one instantaneous output cannot. Future formalization must include state evolution, inputs, intervention, and complete traces. The hidden-factor construction is also deliberately broad. Domain physics, causal structure, conservation laws, geometry, regularity, or minimality may exclude many extensions. The theorem states only that reachable-set equality does not exclude them by itself.

O.3 Composite chains and local loss analysis

The conceptual architecture contains apparatus, sampling, encoding, biological access, internal representation, report, and feedback. The current formal core composes these stages into one map. A richer theory should represent each stage and determine where a distinction is preserved, transformed, compressed, dispersed, or collapsed. Such a theory would permit local refinement of one stage and would show whether a later result contains new information or only a re-encoding of an earlier accessible state.

O.4 The embedded self-interface

The present analyst stands outside the declared map. Wiener, von Foerster, Maturana and Varela, Rosen, von Neumann, and recent computational observer programs supply distinct precedents for treating observation, self-production, and feedback as coupled rather than external operations (Wiener, 1948; von Foerster, 1981; Maturana and Varela, 1980; Rosen, 1991; von Neumann, 1966; Wolfram, 2023; Rickles et al., 2023). The next major formal problem is an embedded system S that contains a self-model and uses the output of a self-interface to update its own dynamics:

$$S_t \xrightarrow{\pi_{\text{self}}} O_t^{\text{self}} \xrightarrow{u} S_{t+1}.$$

The question is not merely whether π_{self} is injective. It is whether the system can justify the interface and update rule using only resources generated inside the same process. Coalgebra, domain theory,

second-order cybernetics, and active inference are candidate tools (von Foerster, 1981; Abramsky et al., 2025; Friston et al., 2025). No general self-justification theorem is claimed here.

O.5 Cross-domain candidate invariant

The candidate invariant should be tested separately in logic, computation, cognition, language, biology, artificial intelligence, and physics:

> *No sufficiently expressive representational system can completely justify or perceive the boundary that generates its own representational capacity.*

A successful program must define *system*, *representation*, *boundary*, *justify*, and *complete* in each domain, then identify a common theorem or explain why the analogy fails. Gödel, Tarski, Lawvere, cybernetics, generalized-observer proposals, and self-referential biology supply test cases rather than proof of universality (Gödel, 1931; Tarski, 1936; Lawvere, 1969; Elshatlawy et al., 2025; Abramsky et al., 2025).

O.6 Paradox as a boundary signal

Hypothesis O.1 (Boundary-signal hypothesis)**.** A recurrent paradox may sometimes mark a representational system encountering the limit of its own closure rather than an ordinary contradiction inside the model.

The hypothesis requires a classifier that distinguishes inconsistent assumptions, semantic ambiguity, formal undecidability, empirical underdetermination, and changed access conditions. It is weakened or rejected if all candidate cases are explained by ordinary error without residual structure. It gains support only if a formally defined boundary measure predicts where otherwise independent paradox-like failures occur.

O.7 Observer-relative mathematics

A further philosophical hypothesis is that mathematics arises through interaction between an observer and reality, while shared mathematics records structures that remain invariant across different representational routes. In its strongest informal form, "what we call mathematics is what observers can preserve and agree upon." This is not implied by the formal theorem. Mathematical structuralism and category-theoretic invariance provide relevant comparison points (Shapiro, 1997; Leinster, 2014). Hilbertian axiomatics separates a structure from one concrete realization, while Newtonian and Einsteinian physics illustrate how stable law or invariant relation can survive changes of coordinates and representation (Hilbert, 1902; Newton, 1999; Einstein, 1905). Feynman's account of physical law supplies a further methodological reminder that mathematical form earns physical force only through contact with experiment (Feynman, 1965). A future paper would need to distinguish conventional notation, structural equivalence, proof-theoretic invariance, and observer-dependent access to mathematical objects.

O.8 Materialism, constraint, and unified-state formation

Hypothesis O.2 (Constraint and unified-state hypothesis)**.** A physical account of consciousness is incomplete unless it explains how high-dimensional activity is constrained into a stable, integrated, reportable regime rather than merely listing the participating components.

This hypothesis is not the claim that matter cannot produce consciousness. It asks which dynamical constraints, integration mechanisms, or model-selection processes make a system behave

as one temporally coherent state. Hard determinist arguments emphasize the causal closure of biological history, while compatibilist accounts locate agency at the level of reasons, alternatives, control, or second-order endorsement (Sapolsky, 2023; List, 2019; Dennett, 2003; Frankfurt, 1971). These positions do not settle consciousness, but they sharpen the requirement that constraint, agency, and reportability be defined at the correct level. Current active-inference, global-workspace, and integrated-information programs offer distinct answers and are now being subjected to prospective comparison (Seth and Bayne, 2022; Cogitate Consortium et al., 2025; Laukkonen et al., 2025; Whyte et al., 2026; Robinson et al., 2025). The interface analysis adds an identifiability question: do the proposed measurements distinguish the relevant internal mechanisms, or do multiple mechanisms remain in one empirical class?

A disconfirming outcome would be a sufficiently complete component-level account that predicts unified reportability without requiring an additional system-level constraint variable. A supporting result would require a defined constraint measure that predicts transitions in reportability across sleep, anesthesia, perturbation, or pathology better than competing explanations.

O.9 Scale-conditioned death and observability

Hypothesis O.3 (Scale-transition hypothesis of death)**.** Death is not annihilation of informational structure but a transition in which that structure is redistributed across scales and degrees of freedom, making it no longer observable as a coherent system at the original scale.

This is a proposed research stance, not a theorem. It must distinguish at least four possibilities: physical persistence of matter and energy, persistence of specific correlations, persistence of organism-level organization, and persistence of subject-level identity. Conservation of energy does not by itself imply conservation of organismal organization or recoverable personal information.

A valid study would define the original-scale coherence criterion, measure selected correlations before and after organismal death, vary spatial and temporal resolution, and determine whether apparent loss at one scale becomes structured signal at another. Failure to detect a structure cannot establish annihilation; detection of a transformed signal cannot establish continuity of the original organism without a reconstruction criterion.

O.10 Light as an informational standard hypothesis

Hypothesis O.4 (Light and living organization)**.** "Light is the informational standard of the living; when it turns into something else it is the informational standard of the dead."

The statement is retained exactly as a named hypothesis because it motivated a proposed measurement route. It is not used as a physical fact. To become testable, *informational standard* must be defined as a measurable statistic or family of statistics, and *turns into something else* must specify energy transfer, spectral change, temporal correlation, spatial dispersion, or coupling to another degree of freedom.

Ultraweak photon emission provides one concrete but limited observable. Living and dead mice show measurable differences in ultraweak emission, and emissions in DNA, cultured cells, and stressed plants can be measured under controlled conditions (Pietruszka and Marzec, 2024; Salari et al., 2025; Mould et al., 2024). These observations neither establish a universal living/dead standard nor show that photon emission encodes complete organismal information. A future study should test whether emission statistics predict defined organization or viability measures beyond temperature, oxidative state, metabolic activity, and tissue composition.

O.11 Single-observer, intentional-forgetting, and external-source hypotheses

The broader metaphysical hypothesis can be stated without converting it into a conclusion: one foundational observer generates a contained reality, relinquishes control or memory, and experiences apparently independent beings through the system. Intentional forgetting is proposed as a condition that permits uncertainty, love, suffering, discovery, and nontrivial choice. A related external-source hypothesis asks whether structure outside the accessible system could leave internal traces through boundary conditions, invariants, or failures of closure.

These hypotheses are neither proved nor refuted by the present theorem. Their rule of admissibility is strict:

No external-source hypothesis earns support without a discriminating prediction.

A changed subjective sense of access, revelation, or certainty is not by itself a discriminating prediction. A valid test must specify an outcome unlikely under the physical, plural-observer, information-only, and cognitive alternatives. If no such outcome can be stated, the hypothesis remains metaphysical and must not be used to explain the formal result.

O.12 Containment, internal reality, and meaning

The interpretive application is that containment does not imply falseness. Experiences and relations may be real within a containing process even when their ontology is dependent. Meaning, beauty, freedom, and truth may be internally stable outputs of a system without being established as absolute outside it. This claim is philosophical rather than empirical. Leibnizian perspectivism, Whiteheadian relation, and Peircean fallibilism provide three different historical comparisons for this claim (Leibniz, 1989; Whitehead, 1978; Peirce, 1877, 1878). Future work should formalize it as invariance under a declared family of transformations: which relations remain stable across observers, interfaces, encodings, or scales?

O.13 Identity, factorial zero, layered tensors, and geometry

A separate mathematical note is reserved for the recurring motif that the unit value 1 appears as $0!$, the empty product, e^0, $\Gamma(1)$, $\det I$, probability normalization, and identity action. The proposed conceptual sequence is

$$\text{preserved unit} \longrightarrow \text{metric structure} \longrightarrow \text{connection} \longrightarrow \text{curvature}.$$

The intended intuition is that Christoffel symbols arise when an identity-like metric structure varies across coordinates, and curvature arises from derivatives and products of the connection. For a specified smooth metric, the Levi-Civita connection depends on the metric and its first derivatives; curvature depends on the connection and its derivatives. Coordinate-dependent metric components can produce nonzero connection coefficients even in flat space, so they do not alone establish curvature. This geometric calculation must be developed independently (Nakahara, 2003). It is not a result of the present theorem, and $0! = 1$ does not by itself generate a connection or curvature tensor.

O.14 Neural manifolds and reportability

Published reconstructions of grid-cell population activity motivate an application of the multi-stage interface problem (Gardner et al., 2022; Hermansen et al., 2024; di Sarra et al., 2025a). The source state, electrode signal, spike-sorted representation, behavioral alignment, dimensional reconstruction, topological summary, biological access, and verbal report are not one object. Future work should define the maps between these levels and test which topological and dynamical conclusions survive justified changes in sampling, preprocessing, reconstruction, and behavioral context.

O.15 Heterogeneous computational and engineered interfaces

The same abstract result should not be assumed to have the same meaning in a digital computer, analog device, quantum processor, brain–computer interface, biological observer, multisensor suite, guidance system, or particle detector. Each system has a different physical state space, encoding, admissible operation, error model, and readout. Future formal work should therefore represent a family of interface implementations

$$(S_i, \eta_i, U_i, \rho_i)$$

and ask when two implementations realize the same abstract map, when they are refinements of one another, and when an output equivalence conceals different internal processes.

Safety-critical systems make the issue concrete. A multisensor suite combines partial, noisy, and sometimes conflicting measurements into an estimated state or decision; the fusion output remains conditional on coverage, calibration, timing, uncertainty, and the selected fusion architecture (Hall and Llinas, 1997; Durrant-Whyte and Henderson, 2008). An inertial guidance system acts from estimated position, velocity, attitude, and uncertainty rather than direct access to an external ground truth (Kalman, 1963; Titterton and Weston, 2004). A particle-physics exclusion contour is conditional on detector acceptance, reconstruction, the tested model, and a statistical rule (Cowan et al., 2011; Particle Data Group, 2026). These cases should be formalized as different interface classes rather than treated as intuitive analogies alone.

O.16 Technical development program

The next mathematical developments are:

(i) stochastic observation channels and equality of induced measures;
(ii) approximate equivalence and quantitative reconstruction error;
(iii) dynamical observability with inputs and interventions;
(iv) composition of multiple interfaces and stage-specific fiber analysis;
(v) information measures for quotient size, dispersion, and refinement cost;
(vi) criteria for minimal or canonical representatives;
(vii) embedded self-interfaces with feedback and model revision;
(viii) automated countermodel and discriminating-experiment construction;
(ix) formal comparison with Leibnizian indiscernibility and perspectival representation;
(x) computational-irreducibility and bounded-observer criteria;
(xi) artificial-system interfaces and machine-consciousness indicator tests;
(xii) translation maps among digital, analog, quantum, neural, and human computational interfaces;
(xiii) state-estimation systems whose actions depend on an internal model and quantified error;
(xiv) multisensor suites that combine partial observations into a bounded state estimate or decision;

(xv) particle-physics acceptance, blind-region, and exclusion analyses as distinct kinds of boundary claim; and
(xvi) domain-specific applications with independently justified maps.

O.17 Hypothesis and application register

Table 12. Status and next requirement for each broader application.

Program	Current status	Required next object	What would count as progress
Embedded boundary invariant	Cross-domain conjecture	Domain-specific definitions and a common translation theorem	A theorem recurring across at least two nontrivially different domains, or a counterexample that narrows the claim
Paradox as boundary signal	Hypothesis	Classifier separating error, undecidability, and empirical underdetermination	Prospective prediction of boundary failures not explained by ordinary inconsistency
Observer-relative mathematics	Philosophical hypothesis	Invariance class across formal languages or observers	A precise theorem connecting translation-invariant structure to interface equivalence
Constraint and unified consciousness	Empirical and computational program	Defined constraint or integration variable and competing models	Preregistered predictions across state transitions that separate mechanisms
Scale-conditioned death	Empirical hypothesis	Coherence criterion across scales and time	Measured transformed structure with a justified continuity or reconstruction relation
Light as informational standard	Not yet operationalized	Spectral, temporal, spatial, or information statistic	Predictive value beyond metabolic, thermal, and oxidative confounds
Single foundational observer	Metaphysical hypothesis	Unique discriminating prediction	An outcome not shared by plural-observer, physical, information-only, or cognitive alternatives
External source or revelation	Metaphysical hypothesis	Publicly checkable boundary condition or invariant	Reproducible evidence with lower likelihood under internal alternatives
Identity-to-geometry motif	Separate mathematical note	Independent differential-geometric derivation	Correct theorem showing the precise relation, or a clear statement of the analogy's limits
Leibnizian perspective and indiscernibility	Historical-formal comparison	Translation between all properties and interface-accessible predicates	A theorem specifying when interface equivalence preserves or fails Leibnizian indiscernibility

Program	Current status	Required next object	What would count as progress
Process and constraint dynamics	Dynamical application program	State evolution, constraints, and measurable order parameters	A model separating process, constraint, and observation effects with falsifiable predictions
Computational irreducibility and observer theory	Comparative formal program	Explicit coarse-graining and resource bounds	A theorem relating computational aggregation to interface refinement or a counterexample separating them
Artificial-system observer interfaces	Empirical and ethical application	Declared outputs, hidden states, self-monitoring channels, and indicator criteria	Tests that distinguish report equivalence from mechanism or consciousness claims
Neural manifold and reportability	Empirical application	Explicit maps from neural activity through measurement and reconstruction to report	Results stable across justified preprocessing, sampling, reconstruction, and behavioral changes
Heterogeneous physical computation	Formal and engineering application	Encoding, evolution, and readout maps for each implementation	A theorem stating when digital, analog, quantum, neural, or human interfaces realize the same abstract relation
Inertial navigation and guidance	State-estimation application	Dynamical model, sensor map, error state, and uncertainty relation	Correct predictions of when distinct physical trajectories remain indistinguishable to the estimator
Multisensor suite and data fusion	Safety-critical state-estimation application	Sensor-specific observation maps, timing, uncertainty, and a declared fusion rule	Correctly calibrated estimates whose blind regions, residuals, and unresolved state components are explicit
Particle-physics forbidden and excluded regions	Experimental application	Separate definitions for theoretical prohibition, detector inaccessibility, and statistical exclusion	Reproducible limits whose dependence on model, acceptance, and confidence rule is explicit

The distinction among theorem, computation, interpretation, empirical application, and metaphysical hypothesis must remain explicit.

P. Motivation and broader questions

P.1 The original epistemological problem

The investigation begins before the usual inverse problem. The original question is not merely whether an unknown state can be reconstructed from data. It is whether a system can fully perceive or justify the boundary that generates its own capacity to represent anything at all. An embedded observer does not stand outside the complete system with unrestricted access. Its body, sensors,

instruments, language, memory, mathematics, and inference rules are themselves parts of the representational route through which the system becomes accessible.

The question has a longer intellectual genealogy than the language of modern inverse problems suggests. Plato's cave separates experienced appearances from the conditions that produce them, while Aristotle distinguishes kinds of cause and levels of explanation (Plato, 1992; Aristotle, 1984). Husserl later asks how scientific objects are constituted relative to the structures of experience and the lifeworld (Husserl, 1970). Leibniz treated each monad as a perspective on one world and coupled that perspectivism to the principles of sufficient reason and the identity of indiscernibles (Leibniz, 1989; Leibniz and Clarke, 2000). Peirce treated inquiry as a fallible process whose concepts acquire content through conceivable practical consequences (Peirce, 1877, 1878). Whitehead shifted emphasis from static substances to processes and relations (Whitehead, 1978). These positions are not adopted as premises. They identify three questions that recur in the present framework: how perspective limits access, what could justify selection among empirically equivalent descriptions, and whether preserved relations may be more stable than a chosen representation.

The strongest candidate invariant motivating the inquiry is therefore:

> *No sufficiently expressive representational system can completely justify or perceive the boundary that generates its own representational capacity.*

This statement is a research conjecture, not a theorem of the present paper. It is adjacent to classical limits on formal self-reference and semantic closure associated with Russell, Gödel, Tarski, and Lawvere (Cantor, 1891; Russell, 1908; Gödel, 1931; Tarski, 1936; Lawvere, 1969; Hofstadter, 1979, 2007); to Kant's distinction between conditions of experience and things considered independently of those conditions (Kant, 1998); and to second-order cybernetics, in which the observer is included within the system of observation (Ashby, 1956; von Foerster, 1981; Maturana and Varela, 1980). Recent work has again called for general theories of observers and formalisms for self-referential living systems (Elshatlawy et al., 2025; Abramsky et al., 2025). None of these sources proves the candidate invariant in the form stated here. They establish the intellectual neighborhood in which it must be tested.

P.2 From the broad conjecture to a bounded theorem

The present paper does not claim to solve the complete embedded-boundary problem. It isolates a necessary mathematical layer. Once a declared source space is mapped into a more limited accessible space, what can be proved about reconstruction, observational equivalence, and model identity from that interface alone?

Let X be a source state space, let O be an accessible observation space, and let $\pi : X \to O$ be the observation map. When

$$\pi(x_1) = \pi(x_2) \qquad \text{for} \qquad x_1 \neq x_2,$$

the two source states are different but indistinguishable through that interface. A procedure whose input is only the shared observation cannot return both states exactly. The analysis develops this local statement into a theory of fibers, quotients, refinement, empirical equivalence, experiments, hidden extensions, and exact decoding.

The bounded theorem is deliberately weaker than the motivating conjecture. It does not prove that every sufficiently expressive system has a noninjective self-interface. It proves what follows *if*

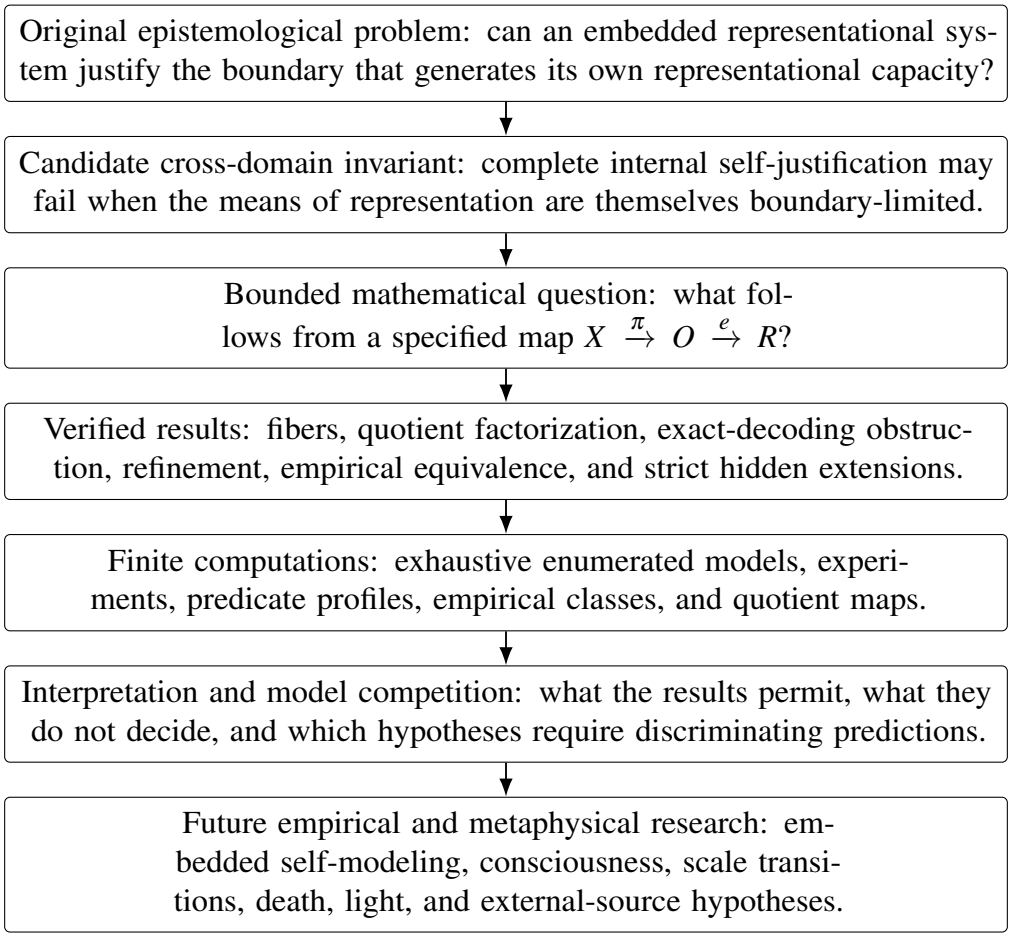


Figure 10. Conceptual hierarchy. The present theorem occupies the verified middle layer. The broader epistemological conjecture and downstream applications are retained but are not presented as consequences of the theorem.

the declared interface is noninjective, and it constructs model extensions showing that equality at an empirical boundary does not force uniqueness of latent representation.

P.3 Three epistemic levels

Every major statement is assigned to one of three levels.

The distinction is essential. Direct experience can motivate a formal question. A theorem can establish a structural limit. Neither step, by itself, proves a preferred metaphysical interpretation. The gap between the second and third levels is not an embarrassment to be concealed. It is the point at which competing models and new measurements are required.

P.4 Competing models and the stopping rule

The motivating problem is not investigated by assuming one total explanation. At minimum, the broader program must compare:

(a) one foundational observer that generates a contained reality and relinquishes access to its own control;
(b) many genuinely independent observers within a shared physical world;
(c) a physical universe that produces bounded biological observers;

Table 13. Epistemic levels used throughout the paper.

Level	Content	Permitted conclusion
Motivating experience or observation	Isolation, self-reference, uncertainty, apparent contradiction, measured outputs, or the felt presence of a boundary.	Establishes the phenomenon or question that requires explanation; does not select an ontology.
Mathematical and computational result	Definitions, deductive theorems, countermodels, finite calculations, and verified code.	Establishes only what follows from stated assumptions and enumerated constructions.
Metaphysical or application hypothesis	A single foundational observer, intentional forgetting, external structure, scale-conditioned death, or a theory of unified consciousness.	Remains a hypothesis until it produces a discriminating prediction and survives empirical comparison.

(d) an information-based reality without a singular foundational observer; and
(e) a cognitive model through which an embedded mind organizes severe uncertainty and self-reference.

These alternatives may agree on all presently accessible observations. The mathematical theory developed in the article explains why such agreement matters: an empirical equivalence class does not select one latent representative without added structure.

The research program therefore adopts a stopping rule:

> *Treat the single-observer model, intentional forgetting, revelation, and every external-source model as metaphysical hypotheses unless they produce a testable prediction that distinguishes them from competing models, including through different outcome probabilities.*

The rule follows Peirce's fallibilist treatment of inquiry and Popper's demand that a scientific proposal expose itself to possible failure rather than acquire protection through reinterpretation (Peirce, 1877; Popper, 1959). It is also consistent with current efforts to make theories of consciousness explicit, preregistered, and vulnerable to divergent outcomes rather than protected by post hoc reinterpretation (Cogitate Consortium et al., 2025; Robinson et al., 2025). It does not abandon the hypotheses. It specifies what would count as progress.

References

Samson Abramsky, Wolfgang Banzhaf, Leo S. D. Caves, Michael Levin, Penousal Machado, Charles Ofria, Susan Stepney, and Roger White. Open questions about time and self-reference in living systems, 2025. URL https://arxiv.org/abs/2508.11423v2. Preprint.

Dorit Aharonov, Jordan Cotler, and Xiao-Liang Qi. Quantum algorithmic measurement. *Nature Communications*, 13:887, 2022. doi: 10.1038/s41467-021-27922-0.

Elizabeth S. Allman, Catherine Matias, and John A. Rhodes. Identifiability of parameters in latent structure models with many observed variables. *The Annals of Statistics*, 37(6A):3099–3132, 2009. doi: 10.1214/09-AOS689.

Animashree Anandkumar, Rong Ge, Daniel Hsu, Sham M. Kakade, and Matus Telgarsky. Tensor decompositions for learning latent variable models. *Journal of Machine Learning Research*, 15: 2773–2832, 2014.

Steen A. Andersson, David Madigan, and Michael D. Perlman. A characterization of markov equivalence classes for acyclic digraphs. *The Annals of Statistics*, 25(2):505–541, 1997. doi: 10.1214/aos/1031833662.

Aristotle. *The Complete Works of Aristotle: The Revised Oxford Translation*. Princeton University Press, 1984. Edited by Jonathan Barnes.

W. Ross Ashby. *An Introduction to Cybernetics*. Chapman & Hall, 1956.

Richard Bellman and Karl Johan Åström. On structural identifiability. *Mathematical Biosciences*, 7 (3–4):329–339, 1970. doi: 10.1016/0025-5564(70)90132-X.

David Blackwell. Comparison of experiments. In Jerzy Neyman, editor, *Proceedings of the Second Berkeley Symposium on Mathematical Statistics and Probability*, pages 93–102, Berkeley, 1951. University of California Press. doi: 10.1525/9780520411586-009.

David Blackwell. Equivalent comparisons of experiments. *The Annals of Mathematical Statistics*, 24(2):265–272, 1953. doi: 10.1214/aoms/1177728248.

David Bohm. *Wholeness and the Implicate Order*. Routledge and Kegan Paul, 1980.

David Bohm and Basil J. Hiley. *The Undivided Universe: An Ontological Interpretation of Quantum Theory*. Routledge, 1993.

Patrick Butlin, Robert Long, Eric Elmoznino, Yoshua Bengio, Jonathan Birch, Axel Constant, George Deane, Stephen M. Fleming, Chris Frith, Xu Ji, Ryota Kanai, Colin Klein, Grace Lindsay, Matthias Michel, Liad Mudrik, Megan A. K. Peters, Eric Schwitzgebel, Jonathan Simon, and Rufin VanRullen. Consciousness in artificial intelligence: Insights from the science of consciousness. *arXiv preprint arXiv:2308.08708*, 2023.

Alberto P. Calderón. On an inverse boundary value problem. *Seminar on Numerical Analysis and Its Applications to Continuum Physics*, pages 65–73, 1980.

Georg Cantor. über eine elementare frage der mannigfaltigkeitslehre. *Jahresbericht der Deutschen Mathematiker-Vereinigung*, 1:75–78, 1891.

David J. Chalmers. Facing up to the problem of consciousness. *Journal of Consciousness Studies*, 2(3):200–219, 1995.

David J. Chalmers. Could a large language model be conscious? *arXiv preprint arXiv:2303.07103*, 2023.

David Maxwell Chickering. A transformational characterization of equivalent bayesian network structures. In *Proceedings of the Eleventh Conference on Uncertainty in Artificial Intelligence*, pages 87–98. Morgan Kaufmann, 1995.

Oana-Teodora Chis, Julio R. Banga, and Eva Balsa-Canto. Structural identifiability of systems biology models: A critical comparison of methods. *PLoS ONE*, 6(11):e27755, 2011. doi: 10.1371/journal.pone.0027755.

Cogitate Consortium, Oscar Ferrante, Urszula Gorska-Klimowska, Simon Henin, Rony Hirschhorn, Aya Khalaf, Alex Lepauvre, et al. Adversarial testing of global neuronal workspace and integrated information theories of consciousness. *Nature*, 642:133–142, 2025. doi: 10.1038/s41586-025-08888-1.

Thomas M. Cover and Joy A. Thomas. *Elements of Information Theory*. Wiley-Interscience, Hoboken, New Jersey, 2 edition, 2006. doi: 10.1002/047174882X.

Glen Cowan, Kyle Cranmer, Eilam Gross, and Ofer Vitells. Asymptotic formulae for likelihood-based tests of new physics. *The European Physical Journal C*, 71:1554, 2011. doi: 10.1140/epjc/s10052-011-1554-0.

Leonardo de Moura, Soonho Kong, Jeremy Avigad, Floris van Doorn, and Jakob von Raumer. The lean theorem prover (system description). In *Automated Deduction – CADE-25*, volume 9195 of *Lecture Notes in Computer Science*, pages 378–388. Springer, 2015. doi: 10.1007/978-3-319-21401-6_26.

Terrence W. Deacon. *Incomplete Nature: How Mind Emerged from Matter*. W. W. Norton, 2011.

Daniel C. Dennett. *Freedom Evolves*. Viking, 2003.

David Deutsch. Constructor theory. *Synthese*, 190:4331–4359, 2013. doi: 10.1007/s11229-013-0279-z.

David Deutsch and Chiara Marletto. Constructor theory of information. *Proceedings of the Royal Society A*, 471(2174):20140540, 2015. doi: 10.1098/rspa.2014.0540.

Giovanni di Sarra, Siddharth Jha, and Yasser Roudi. The role of oscillations in grid cells' toroidal topology. *PLOS Computational Biology*, 21(1):e1012776, 2025a. doi: 10.1371/journal.pcbi.1012776.

Giovanni di Sarra, Siddharth Jha, and Yasser Roudi. Correction: The role of oscillations in grid cells' toroidal topology. *PLOS Computational Biology*, 21(10):e1013623, 2025b. doi: 10.1371/journal.pcbi.1013623.

David P. DiVincenzo. The physical implementation of quantum computation. *Fortschritte der Physik*, 48(9–11):771–783, 2000. doi: 10.1002/1521-3978(200009)48:9/11<771::AID-PROP771>3.0.CO;2-E.

Hugh Durrant-Whyte and Thomas C. Henderson. Multisensor data fusion. In Bruno Siciliano and Oussama Khatib, editors, *Springer Handbook of Robotics*, pages 585–610. Springer, Berlin, Heidelberg, 2008. doi: 10.1007/978-3-540-30301-5_26.

Albert Einstein. On the electrodynamics of moving bodies. *Annalen der Physik*, 17:891–921, 1905.

Hatem Elshatlawy, Dean Rickles, Xerxes D. Arsiwalla, and Alexander Blum. Towards a generalized theory of observers, 2025. URL https://arxiv.org/abs/2504.16225v1. Preprint.

Heinz W. Engl, Martin Hanke, and Andreas Neubauer. *Regularization of Inverse Problems*. Kluwer Academic Publishers, Dordrecht, 1996. doi: 10.1007/978-94-009-1740-8.

Richard P. Feynman. *The Character of Physical Law*. MIT Press, 1965.

Harry G. Frankfurt. Freedom of the will and the concept of a person. *The Journal of Philosophy*, 68 (1):5–20, 1971. doi: 10.2307/2024717.

Karl Friston. The free-energy principle: A unified brain theory? *Nature Reviews Neuroscience*, 11 (2):127–138, 2010. doi: 10.1038/nrn2787.

Karl Friston, Conor Heins, Tim Verbelen, Lancelot Da Costa, Tommaso Salvatori, Dimitrije Markovic, Alexander Tschantz, Magnus Koudahl, Christopher Buckley, and Thomas Parr. From pixels to planning: Scale-free active inference. *Frontiers in Network Physiology*, 5:1521963, 2025. doi: 10.3389/fnetp.2025.1521963.

Richard J. Gardner, Erik Hermansen, Marius Pachitariu, Yoram Burak, Nils A. Baas, Benjamin A. Dunn, May-Britt Moser, and Edvard I. Moser. Toroidal topology of population activity in grid cells. *Nature*, 602(7895):123–128, 2022. doi: 10.1038/s41586-021-04268-7.

Kurt Gödel. Über formal unentscheidbare sätze der principia mathematica und verwandter systeme i. *Monatshefte für Mathematik und Physik*, 38:173–198, 1931. doi: 10.1007/BF01700692.

Alex Gomez-Marin and Anil K. Seth. A science of consciousness beyond pseudo-science and pseudo-consciousness. *Nature Neuroscience*, 28:703–706, 2025. doi: 10.1038/s41593-025-01913-6.

Michael S. A. Graziano. *Consciousness and the Social Brain*. Oxford University Press, 2013.

Luigi Gresele, Paul K. Rubenstein, Arash Mehrjou, Francesco Locatello, and Bernhard Schölkopf. The incomplete rosetta stone problem: Identifiability results for multi-view nonlinear ica. In *Proceedings of the 35th Uncertainty in Artificial Intelligence Conference*, volume 115 of *Proceedings of Machine Learning Research*, pages 217–227. PMLR, 2020.

Stephen Grossberg. *Conscious Mind, Resonant Brain: How Each Brain Makes a Mind*. Oxford University Press, 2021.

Joseph H. A. Guillaume, John D. Jakeman, Stefano Marsili-Libelli, Michael Asher, Philip Brunner, Barry Croke, Mary C. Hill, Anthony J. Jakeman, Karel J. Keesman, Saman Razavi, and Johannes D. Stigter. Introductory overview of identifiability analysis: A guide to evaluating whether you have the right type of data for your modeling purpose. *Environmental Modelling & Software*, 119:418–432, 2019. doi: 10.1016/j.envsoft.2019.07.007.

Jacques Hadamard. *Lectures on Cauchy's Problem in Linear Partial Differential Equations*. Yale University Press, New Haven, 1923.

Thomas Hales, Mark Adams, Gertrud Bauer, Tat Dat Dang, John Harrison, Le Truong Hoang, Cezary Kaliszyk, Victor Magron, Sean McLaughlin, Tat Thang Nguyen, Quang Truong Nguyen, Tobias Nipkow, Steven Obua, Joseph Pleso, Jason Rute, Alexey Solovyev, Thi Hoai An Ta, Nam Trung Tran, Thi Diep Trieu, Josef Urban, Ky Vu, and Roland Zumkeller. A formal proof of the kepler conjecture. *Forum of Mathematics, Pi*, 5:e2, 2017. doi: 10.1017/fmp.2017.1.

David L. Hall and James Llinas. An introduction to multisensor data fusion. *Proceedings of the IEEE*, 85(1):6–23, 1997. doi: 10.1109/5.554205.

Paul R. Halmos. *Naive Set Theory*. Undergraduate Texts in Mathematics. Springer, 1974. doi: 10.1007/978-1-4757-1645-0.

Alain Hauser and Peter Bühlmann. Characterization and greedy learning of interventional markov equivalence classes of directed acyclic graphs. *Journal of Machine Learning Research*, 13: 2409–2464, 2012.

Robert Hermann and Arthur J. Krener. Nonlinear controllability and observability. *IEEE Transactions on Automatic Control*, 22(5):728–740, 1977. doi: 10.1109/TAC.1977.1101601.

Erik Hermansen, David A. Klindt, and Benjamin A. Dunn. Uncovering 2-d toroidal representations in grid cell ensemble activity during 1-d behavior. *Nature Communications*, 15:5429, 2024. doi: 10.1038/s41467-024-49703-1.

David Hilbert. *The Foundations of Geometry*. Open Court, 1902.

Donald D. Hoffman and Chetan Prakash. Objects of consciousness. *Frontiers in Psychology*, 5:577, 2014. doi: 10.3389/fpsyg.2014.00577.

Douglas R. Hofstadter. *Gödel, Escher, Bach: An Eternal Golden Braid*. Basic Books, 1979.

Douglas R. Hofstadter. *I Am a Strange Loop*. Basic Books, 2007.

Dominic Horsman, Susan Stepney, Rob C. Wagner, and Viv Kendon. When does a physical system compute? *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 470(2169):20140182, 2014. doi: 10.1098/rspa.2014.0182.

Dominic C. Horsman. Abstraction/representation theory for heterotic physical computing. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 373(2046):20140240, 2015. doi: 10.1098/rsta.2014.0240.

Edmund Husserl. *The Crisis of European Sciences and Transcendental Phenomenology*. Northwestern University Press, 1970.

Aapo Hyvärinen, Hiroaki Sasaki, and Richard Turner. Nonlinear ica using auxiliary variables and generalized contrastive learning. In *Proceedings of the Twenty-Second International Conference on Artificial Intelligence and Statistics*, volume 89 of *Proceedings of Machine Learning Research*, pages 859–868. PMLR, 2019.

Julian Jaynes. *The Origin of Consciousness in the Breakdown of the Bicameral Mind*. Houghton Mifflin, 1976.

Rudolf E. Kalman. Mathematical description of linear dynamical systems. *Journal of the Society for Industrial and Applied Mathematics, Series A: Control*, 1(2):152–192, 1963. doi: 10.1137/0301010.

Immanuel Kant. *Critique of Pure Reason*. Cambridge University Press, 1998. Original editions 1781 and 1787; Edited by Paul Guyer, Allen W. Wood.

Stuart A. Kauffman. *The Origins of Order: Self-Organization and Selection in Evolution*. Oxford University Press, 1993.

Ilyes Khemakhem, Diederik Kingma, Ricardo Monti, and Aapo Hyvärinen. Variational autoencoders and nonlinear ica: A unifying framework. In *Proceedings of the Twenty Third International Conference on Artificial Intelligence and Statistics*, volume 108 of *Proceedings of Machine Learning Research*, pages 2207–2217. PMLR, 2020.

Christof Koch. *The Feeling of Life Itself: Why Consciousness Is Widespread but Can't Be Computed*. MIT Press, 2019.

Joseph B. Kruskal. Three-way arrays: Rank and uniqueness of trilinear decompositions, with application to arithmetic complexity and statistics. *Linear Algebra and its Applications*, 18(2):95–138, 1977. doi: 10.1016/0024-3795(77)90069-6.

Solomon Kullback and Richard A. Leibler. On information and sufficiency. *The Annals of Mathematical Statistics*, 22(1):79–86, 1951. doi: 10.1214/aoms/1177729694.

Kim G. Larsen and Arne Skou. Bisimulation through probabilistic testing. *Information and Computation*, 94(1):1–28, 1991. doi: 10.1016/0890-5401(91)90030-6.

Ruben Laukkonen, Karl Friston, and Shamil Chandaria. A beautiful loop: An active inference theory of consciousness. *Neuroscience & Biobehavioral Reviews*, 176:106296, 2025. doi: 10.1016/j.neubiorev.2025.106296.

F. William Lawvere. Diagonal arguments and cartesian closed categories. In *Category Theory, Homology Theory and Their Applications II*, volume 92 of *Lecture Notes in Mathematics*, pages 134–145. Springer, 1969. doi: 10.1007/BFb0080769.

Lucien Le Cam. *Asymptotic Methods in Statistical Decision Theory*. Springer, New York, 1986. doi: 10.1007/978-1-4612-4946-7.

Lucien Le Cam and Grace Lo Yang. *Asymptotics in Statistics: Some Basic Concepts*. Springer, New York, 2 edition, 2000.

Gottfried Wilhelm Leibniz. *Philosophical Essays*. Hackett Publishing Company, 1989. Edited by Roger Ariew, Daniel Garber.

Gottfried Wilhelm Leibniz and Samuel Clarke. *G. W. Leibniz and Samuel Clarke: Correspondence*. Hackett Publishing Company, 2000. Edited by Roger Ariew.

Tom Leinster. *Basic Category Theory*, volume 143 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, 2014. doi: 10.1017/CBO9781107360068.

Christian List. *Why Free Will Is Real*. Harvard University Press, 2019.

Francesco Locatello, Stefan Bauer, Mario Lucic, Gunnar Raetsch, Sylvain Gelly, Bernhard Schölkopf, and Olivier Bachem. Challenging common assumptions in the unsupervised learning of disentangled representations. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 4114–4124. PMLR, 2019.

Saunders Mac Lane and Garrett Birkhoff. *Algebra*, volume 330 of *AMS Chelsea Publishing*. American Mathematical Society, 3 edition, 1999. ISBN 9780821816462.

Humberto R. Maturana and Francisco J. Varela. *Autopoiesis and Cognition: The Realization of the Living*, volume 42 of *Boston Studies in the Philosophy and History of Science*. D. Reidel, 1980. doi: 10.1007/978-94-009-8947-4.

Nicolette Meshkat, Marisa Eisenberg, and Joseph J. DiStefano. An algorithm for finding globally identifiable parameter combinations of nonlinear ode models using gröbner bases. *Mathematical Biosciences*, 222(2):61–72, 2009. doi: 10.1016/j.mbs.2009.08.010.

Robin Milner. *Communication and Concurrency*. Prentice Hall, New York, 1989.

Melanie Mitchell. *Complexity: A Guided Tour*. Oxford University Press, 2009.

Edward F. Moore. Gedanken-experiments on sequential machines. In Claude E. Shannon and John McCarthy, editors, *Automata Studies*, volume 34 of *Annals of Mathematics Studies*, pages 129–153. Princeton University Press, Princeton, 1956.

Rhys R. Mould, Alasdair M. Mackenzie, Ifigeneia Kalampouka, Alistair V. W. Nunn, E. Louise Thomas, Jimmy D. Bell, and Stanley W. Botchway. Ultra weak photon emission—a brief review. *Frontiers in Physiology*, 15:1348915, 2024. doi: 10.3389/fphys.2024.1348915.

Mikio Nakahara. *Geometry, Topology and Physics*. CRC Press, 2 edition, 2003.

Isaac Newton. *The Principia: Mathematical Principles of Natural Philosophy*. University of California Press, 1999.

Gregoire Nicolis and Ilya Prigogine. *Self-Organization in Nonequilibrium Systems: From Dissipative Structures to Order through Fluctuations*. Wiley, 1977.

David M. R. Park. Concurrency and automata on infinite sequences. In *Theoretical Computer Science: 5th GI-Conference*, volume 104 of *Lecture Notes in Computer Science*, pages 167–183. Springer, 1981. doi: 10.1007/BFb0017309.

Thomas Parr, Giovanni Pezzulo, and Karl J. Friston. *Active Inference: The Free Energy Principle in Mind, Brain, and Behavior*. MIT Press, 2022.

Particle Data Group. Review of particle physics. *International Journal of Modern Physics A*, 41: 2630011, 2026. URL https://pdg.lbl.gov/2026/.

Judea Pearl. *Causality: Models, Reasoning, and Inference*. Cambridge University Press, Cambridge, 2 edition, 2009.

Charles Sanders Peirce. The fixation of belief. *Popular Science Monthly*, 12:1–15, 1877.

Charles Sanders Peirce. How to make our ideas clear. *Popular Science Monthly*, 12:286–302, 1878.

Roger Penrose. *The Emperor's New Mind: Concerning Computers, Minds, and the Laws of Physics*. Oxford University Press, 1989.

Roger Penrose. *Shadows of the Mind: A Search for the Missing Science of Consciousness*. Oxford University Press, 1994.

Mariusz Pietruszka and Marek Marzec. Ultra-weak photon emission from dna. *Scientific Reports*, 14:28915, 2024. doi: 10.1038/s41598-024-80469-0.

Plato. *Republic*. Hackett Publishing Company, 1992. Edited by C. D. C. Reeve.

Karl R. Popper. *The Logic of Scientific Discovery*. Hutchinson, 1959.

Robert Prentner and Donald D. Hoffman. Interfacing consciousness. *Frontiers in Psychology*, 15: 1429376, 2024. doi: 10.3389/fpsyg.2024.1429376.

Rajesh P. N. Rao and Dana H. Ballard. Predictive coding in the visual cortex: A functional interpretation of some extra-classical receptive-field effects. *Nature Neuroscience*, 2(1):79–87, 1999. doi: 10.1038/4580.

Andreas Raue, Clemens Kreutz, Thomas Maiwald, Julie Bachmann, Marcel Schilling, Ursula Klingmüller, and Jens Timmer. Structural and practical identifiability analysis of partially observed dynamical models by exploiting the profile likelihood. *Bioinformatics*, 25(15):1923–1929, 2009. doi: 10.1093/bioinformatics/btp358.

Dean Rickles, Hussein Elshatlawy, and Xerxes D. Arsiwalla. Ruliology: Linking computation, observers and physical law. *arXiv preprint arXiv:2308.16068*, 2023.

Jonathan Edward Robinson, Andrew W. Corcoran, Christopher J. Whyte, András Sárközy, Anil K. Seth, Gyula Kovács, Karl J. Friston, Cyriel M. A. Pennartz, Giulio Tononi, and Jakob Hohwy. The role of active inference in conscious awareness. *PLOS ONE*, 20(12):e0328836, 2025. doi: 10.1371/journal.pone.0328836.

Robert Rosen. *Life Itself: A Comprehensive Inquiry into the Nature, Origin, and Fabrication of Life*. Columbia University Press, 1991.

Thomas J. Rothenberg. Identification in parametric models. *Econometrica*, 39(3):577–591, 1971. doi: 10.2307/1913267.

Bertrand Russell. Mathematical logic as based on the theory of types. *American Journal of Mathematics*, 30(3):222–262, 1908. doi: 10.2307/2369948.

Jan J. M. M. Rutten. Universal coalgebra: A theory of systems. *Theoretical Computer Science*, 249 (1):3–80, 2000. doi: 10.1016/S0304-3975(00)00056-6.

V. Salari, V. Seshan, L. Frankle, D. England, C. Simon, and D. Oblak. Imaging ultraweak photon emission from living and dead mice and from plants under stress. *The Journal of Physical Chemistry Letters*, 16(17):4354–4362, 2025. doi: 10.1021/acs.jpclett.4c03546.

Robert M. Sapolsky. *Determined: A Science of Life without Free Will*. Penguin Press, 2023.

Anil K. Seth and Tim Bayne. Theories of consciousness. *Nature Reviews Neuroscience*, 23:439–452, 2022. doi: 10.1038/s41583-022-00587-4.

Claude E. Shannon. Mathematical theory of the differential analyzer. *Journal of Mathematics and Physics*, 20(1–4):337–354, 1941. doi: 10.1002/sapm1941201337.

Claude E. Shannon. A mathematical theory of communication, part i. *Bell System Technical Journal*, 27(3):379–423, 1948a. doi: 10.1002/j.1538-7305.1948.tb01338.x.

Claude E. Shannon. A mathematical theory of communication, part ii. *Bell System Technical Journal*, 27(4):623–656, 1948b. doi: 10.1002/j.1538-7305.1948.tb00917.x.

Stewart Shapiro. *Philosophy of Mathematics: Structure and Ontology*. Oxford University Press, 1997.

Eduardo D. Sontag. *Mathematical Control Theory: Deterministic Finite Dimensional Systems*, volume 6 of *Texts in Applied Mathematics*. Springer, New York, 2 edition, 1998. doi: 10.1007/978-1-4612-0577-7.

Alfred Tarski. Der wahrheitsbegriff in den formalisierten sprachen. *Studia Philosophica*, 1:261–405, 1936.

Max Tegmark. Consciousness as a state of matter. *Chaos, Solitons and Fractals*, 76:238–270, 2015. doi: 10.1016/j.chaos.2015.03.014.

Henry Teicher. Identifiability of finite mixtures. *The Annals of Mathematical Statistics*, 34(4): 1265–1269, 1963. doi: 10.1214/aoms/1177703862.

Joshua B. Tenenbaum, Charles Kemp, Thomas L. Griffiths, and Noah D. Goodman. How to grow a mind: Statistics, structure, and abstraction. *Science*, 331(6022):1279–1285, 2011. doi: 10.1126/science.1192788.

The mathlib Community. The lean mathematical library. In *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs*, pages 367–381. ACM, 2020. doi: 10.1145/3372885.3373824.

Andrey N. Tikhonov. Solution of incorrectly formulated problems and the regularization method. *Soviet Mathematics Doklady*, 4:1035–1038, 1963.

David H. Titterton and John L. Weston. *Strapdown Inertial Navigation Technology*. Institution of Engineering and Technology, 2 edition, 2004. doi: 10.1049/PBRA017E.

Giulio Tononi. Consciousness as integrated information: A provisional manifesto. *The Biological Bulletin*, 215(3):216–242, 2008. doi: 10.2307/25470707.

Giulio Tononi, Melanie Boly, Marcello Massimini, and Christof Koch. Consciousness or pseudo-consciousness? a clash of two paradigms. *Nature Neuroscience*, 28:694–702, 2025. doi: 10.1038/s41593-025-01880-y.

Erik Torgersen. *Comparison of Statistical Experiments*. Cambridge University Press, Cambridge, 1991. doi: 10.1017/CBO9780511666353.

Thomas Verma and Judea Pearl. Equivalence and synthesis of causal models. In *Proceedings of the Sixth Annual Conference on Uncertainty in Artificial Intelligence*, pages 255–270. Elsevier Science, 1990.

Alejandro F. Villaverde, Antonio Barreiro, and Antonis Papachristodoulou. Structural identifiability of dynamic systems biology models. *PLoS Computational Biology*, 12(10):e1005153, 2016. doi: 10.1371/journal.pcbi.1005153.

Heinz von Foerster. *Observing Systems*. Intersystems Publications, 1981.

John von Neumann. *Mathematical Foundations of Quantum Mechanics*. Princeton University Press, 1955.

John von Neumann. *Theory of Self-Reproducing Automata*. University of Illinois Press, 1966. Edited by Arthur W. Burks.

Eric Walter and Luc Pronzato. *Identification of Parametric Models from Experimental Data*. Communications and Control Engineering. Springer, Berlin, 1997.

John Archibald Wheeler. Law without law. In John Archibald Wheeler and Wojciech H. Zurek, editors, *Quantum Theory and Measurement*, pages 182–213. Princeton University Press, 1983.

Alfred North Whitehead. *Process and Reality: An Essay in Cosmology*. Free Press, corrected edition, 1978. Edited by David Ray Griffin, Donald W. Sherburne.

Christopher J. Whyte, Andrew W. Corcoran, Jonathan Robinson, Ryan Smith, Rosalyn J. Moran, Thomas Parr, Karl J. Friston, Anil K. Seth, and Jakob Hohwy. On the minimal theory of consciousness implicit in active inference. *Physics of Life Reviews*, 56:4–28, 2026. doi: 10.1016/j.plrev.2025.11.002.

Norbert Wiener. *Cybernetics: Or Control and Communication in the Animal and the Machine*. MIT Press, 1948.

Francis R. Willett, Erin M. Kunz, Chaofei Fan, Donald T. Avansino, Guy H. Wilson, Eun Young Choi, Fahad Kamdar, Matthew F. Glasser, Leigh R. Hochberg, Shaul Druckmann, Krishna V. Shenoy, and Jaimie M. Henderson. A high-performance speech neuroprosthesis. *Nature*, 620: 1031–1036, 2023. doi: 10.1038/s41586-023-06377-x.

Stephen Wolfram. *A New Kind of Science*. Wolfram Media, 2002.

Stephen Wolfram. Observer theory, 2023. URL https://www.wolframphysics.org/technical-introduction/observer-theory/.

Jonathan R. Wolpaw, Niels Birbaumer, Dennis J. McFarland, Gert Pfurtscheller, and Theresa M. Vaughan. Brain–computer interfaces for communication and control. *Clinical Neurophysiology*, 113(6):767–791, 2002. doi: 10.1016/S1388-2457(02)00057-3.

Quanhan Xi and Benjamin Bloem-Reddy. Indeterminacy in generative models: Characterization and strong identifiability. In *Proceedings of the 26th International Conference on Artificial Intelligence and Statistics*, volume 206 of *Proceedings of Machine Learning Research*, pages 6912–6939. PMLR, 2023.